

Vlasov theory of warm electrostatic waves in a magnetized plasma

8.1 Solving the Vlasov equation by tracking each particle's history

It has been tacitly assumed until now that the wave phase experienced by a particle is the phase the particle would have experienced if it had not deviated from its initial position $\mathbf{x}_0$. This means that the particle trajectory used when determining the wave phase experienced by the particle is effectively assumed to be $\mathbf{x} = \mathbf{x}_0$ instead of the actual trajectory $\mathbf{x} = \mathbf{x}(t)$. Thus, the wave phase seen by the particle was approximated as

$$\mathbf{k} \cdot \mathbf{x}(t) - \omega t \simeq \mathbf{k} \cdot \mathbf{x}_0 - \omega t. \quad (8.1)$$

This approximation is fine provided the deviation of the actual trajectory from the assumed trajectory satisfies the condition

$$|\mathbf{k} \cdot (\mathbf{x}(t) - \mathbf{x}_0)| \ll \pi/2, \quad (8.2)$$

in which case phase error due to the deviation would be insignificant. Two situations can occur where this condition is not satisfied, namely:

1. the wave amplitude is so large that the particle displacement due to the wave is significant compared to a wavelength,
2. the wave amplitude is small, but the particle has a large initial velocity so it moves substantially during a wave period.

The first case results in chaotic particle motion as discussed in Section 3.7.3 while the second case, the subject of this chapter, occurs when particles have significant thermal motion. If the motion is parallel to the magnetic field, significant thermal motion means that v_T is non-negligible compared to $\omega/k_{\parallel}$, a regime already discussed in Section 5.3 for unmagnetized plasmas. The subscript $\parallel$ is used here to denote the direction along the magnetic field. If the magnetic field is straight and given by $\mathbf{B} = B\hat{z}$, $\parallel$ would simply be the z direction, but in a more general situation the $\parallel$ component of a vector would be obtained by dotting the

vector with $\hat{B}$. If the motion is perpendicular to the magnetic field (this direction is denoted as $\perp$), then significant thermal motion means that $k_{\perp} r_L$ becomes large, i.e., the Larmor orbit r_L becomes comparable to the wavelength. In the situation where $k_{\perp} r_L$ is not infinitesimal, the particle samples a continuum of wave phases as it traces out its Larmor orbit and these sampled wave phases are considerably different from the wave phase at the particle's gyrocenter.

Consider an electrostatic wave with potential

$$\phi_1(\mathbf{x}, t) = \tilde{\phi}_1 \exp(i\mathbf{k} \cdot \mathbf{x} - i\omega t). \quad (8.3)$$

As before, the convention will be used that a tilde refers to the amplitude of a perturbed quantity; if there is no tilde, then the exponential phase factor is understood to be included. Because the wave is electrostatic, Poisson's equation is the relevant equation relating particle motion to the electric field and is

$$k^2 \phi_1 = \frac{1}{\epsilon_0} \sum_{\sigma} n_{\sigma 1} q_{\sigma}, \quad (8.4)$$

where $n_{\sigma 1}$ is the density perturbation for each species σ . Since the density perturbation is just the zeroth moment of the perturbed distribution function,

$$n_{\sigma 1} = \int f_{\sigma 1} d^3v, \quad (8.5)$$

the problem reduces to determining the perturbed distribution function $f_{\sigma 1}$.

In the presence of a uniform magnetic field the linearized Vlasov equation is

$$\frac{\partial f_{\sigma 1}}{\partial t} + \mathbf{v} \cdot \frac{\partial f_{\sigma 1}}{\partial \mathbf{x}} + \frac{q_{\sigma}}{m_{\sigma}} (\mathbf{v} \times \mathbf{B}) \cdot \frac{\partial f_{\sigma 1}}{\partial \mathbf{v}} = \frac{q_{\sigma}}{m_{\sigma}} \nabla \phi_1 \cdot \frac{\partial f_{\sigma 0}}{\partial \mathbf{v}}, \quad (8.6)$$

where the subscript 0 refers to equilibrium quantities and the subscript 1 to first-order perturbations (no 0 subscript has been used to distinguish the equilibrium magnetic field $\mathbf{B}$ from a perturbed magnetic field because, being electrostatic, the assumed wave has no associated perturbed magnetic field).

Consider all particles of species σ located at an arbitrary point $\mathbf{x}, \mathbf{v}$ in phase-space at the present time t . These particles must all have identical phase-space trajectories in both the future and the past because they are all acted on by the same Lorentz force $q_{\sigma}(\mathbf{E}(\mathbf{x}, t) + \mathbf{v} \times \mathbf{B}(\mathbf{x}, t))$ and they all have the same initial condition at time t since, by assumption, they have the same $\mathbf{x}$ and $\mathbf{v}$ at time t . By integrating the equation of motion starting from this point in phase-space, the phase-space trajectory $\mathbf{x}(t'), \mathbf{v}(t')$ of this set of particles can be determined. The boundary conditions on such a phase-space trajectory are simply

$$\mathbf{x}(t) = \mathbf{x}, \quad \mathbf{v}(t) = \mathbf{v}. \quad (8.7)$$

Instead of treating $\mathbf{x}, \mathbf{v}$ as independent variables denoting a point in phase-space, let us think of these quantities as temporal boundary conditions for particles with

phase-space trajectories $\mathbf{x}(t)$, $\mathbf{v}(t)$ that happen to be at location $\mathbf{x}$, $\mathbf{v}$ at time t . Thus, the velocity distribution function for all particles that happen to be at phase-space location $\mathbf{x}$, $\mathbf{v}$ at time t is $f_{\sigma 1} = f_{\sigma 1}(\mathbf{x}(t), \mathbf{v}(t), t)$ and since $\mathbf{x}$ and $\mathbf{v}$ were arbitrary, this expression is valid at any point in phase-space. The time derivative of this function is

$$\frac{d}{dt} f_{\sigma 1}(\mathbf{x}(t), \mathbf{v}(t), t) = \frac{\partial f_{\sigma 1}}{\partial t} + \frac{\partial f_{\sigma 1}}{\partial \mathbf{x}} \cdot \frac{d\mathbf{x}}{dt} + \frac{\partial f_{\sigma 1}}{\partial \mathbf{v}} \cdot \frac{d\mathbf{v}}{dt}. \quad (8.8)$$

In principle, the wave force acting on the particles ought to be taken into account when calculating their trajectories. However, if the wave amplitude is sufficiently small, the particle trajectory will not be significantly affected by the wave and so will be essentially the same as the unperturbed trajectory, namely the trajectory the particle would have had if there were no wave. Since the unperturbed particle trajectory equations are

$$\frac{d\mathbf{x}}{dt} = \mathbf{v}, \quad \frac{d\mathbf{v}}{dt} = \frac{q_{\sigma}}{m_{\sigma}} \mathbf{v} \times \mathbf{B}, \quad (8.9)$$

it is seen that Eq. (8.8) is identical to the left-hand side of Eq. (8.6). Equation (8.6) can thus be rewritten as

$$\left(\frac{d}{dt} f_{\sigma 1}(\mathbf{x}(t), \mathbf{v}(t), t) \right)_{\text{unperturbed trajectory}} = \frac{q_{\sigma}}{m_{\sigma}} \nabla \phi_1 \cdot \frac{\partial f_{\sigma 0}}{\partial \mathbf{v}}, \quad (8.10)$$

where the left-hand side is the derivative of the distribution function that would be measured by an observer sitting on a particle having the *unperturbed* phase-space trajectory $\mathbf{x}(t)$, $\mathbf{v}(t)$. Equation (8.10) may be integrated to give

$$f_{\sigma 1}(\mathbf{x}, \mathbf{v}, t) = \frac{q_{\sigma}}{m_{\sigma}} \int_{-\infty}^t dt' \left[\nabla \phi_1 \cdot \frac{\partial f_{\sigma 0}}{\partial \mathbf{v}} \right]_{\mathbf{x}=\mathbf{x}(t'), \mathbf{v}=\mathbf{v}(t')}. \quad (8.11)$$

If the right-hand side of Eq. (8.10) is considered as a “force” acting to change the perturbed distribution function, then Eq. (8.11) is effectively a statement that the perturbed distribution function at $\mathbf{x}$, $\mathbf{v}$ for time t is a result of the sum of all the “forces” acting over times prior to t evaluated along the unperturbed trajectory of the particle. In effect, Eq. (8.11) states that, just as the present state of a person is in a sense the consequence of the cumulative effect of all the various influences experienced by the person at different times and places during his/her entire previous life, the present status of the particles, denoted by $f_{\sigma 1}(\mathbf{x}, \mathbf{v}, t)$, is the consequence of the cumulative effect of all the various influences (i.e., forces) experienced at different times and places by the particles during their entire past history. “Unperturbed trajectories” refers to the solution to Eqs. (8.9); these equations neglect any wave-induced changes to the particle trajectory and

simply give the trajectory of a thermal particle. The “force” in Eq. (8.11) must be evaluated along the *past* phase-space trajectory because *that* is where the particles at $\mathbf{x}, \mathbf{v}$ were located at previous times and so that is where the particles “felt” the “force.” This is called “integrating along the *unperturbed* orbits” and is valid only when the unperturbed orbits (trajectories) are a good approximation to the particles’ actual orbits. Mathematically speaking, these unperturbed orbits are the characteristics of the left-hand side of Eq. (8.6), a homogeneous hyperbolic partial differential equation. The solutions of this homogeneous equation are constant along the characteristics. The right-hand side is the inhomogeneous or forcing term and acts to modify the homogeneous solution; the cumulative effect of this force is found by integrating along the characteristics of the homogeneous part.

The problem is now formally solved; all that is required is an explicit evaluation of the integrals. The functional form of the equilibrium distribution function is determined by the specific physical problem under consideration. Often the plasma has a uniform Maxwellian distribution

$$f_{\sigma 0}(\mathbf{v}) = \frac{n_{\sigma 0}}{\pi^{3/2} v_{T\sigma}^3} e^{-v^2/v_{T\sigma}^2}, \quad (8.12)$$

where

$$v_{T\sigma} = \sqrt{2\kappa T_{\sigma}/m_{\sigma}}. \quad (8.13)$$

It must be understood that Eq. (8.12) represents one of an infinite number of possible choices for the equilibrium distribution function – *any* other function of the constants of the motion would also be valid. In fact, functions differing from Eq. (8.12) will later be used to model drifting plasmas and plasmas with density gradients.

Substitution of Eq. (8.12) into the orbit integral Eq. (8.11) gives

$$\begin{aligned} f_{\sigma 1}(\mathbf{x}, \mathbf{v}, t) = & -\frac{2n_{\sigma 0}q_{\sigma}\tilde{\phi}_1}{\pi^{3/2}m_{\sigma}v_{T\sigma}^3} \exp[-v^2/v_{T\sigma}^2] \\ & \times \int_{-\infty}^t dt' \left\{ \frac{i\mathbf{k} \cdot \mathbf{v}(t')}{v_{T\sigma}^2} \exp[i\mathbf{k} \cdot \mathbf{x}(t') - i\omega t'] \right\}. \end{aligned} \quad (8.14)$$

Since the kinetic energy $mv^2/2$ of the unperturbed orbits is a constant of the motion, the quantity $\exp[-v^2/v_{T\sigma}^2]$ has been factored out of the time integral. This orbit integral may be simplified by noting

$$\frac{d}{dt'} \exp[i\mathbf{k} \cdot \mathbf{x}(t')] = i\mathbf{k} \cdot \mathbf{v}(t') \exp[i\mathbf{k} \cdot \mathbf{x}(t')] \quad (8.15)$$

so Eq. (8.14) becomes

$$\begin{aligned} f_{\sigma 1}(\mathbf{x}, \mathbf{v}, t) &= -\frac{q_{\sigma}}{\kappa T_{\sigma}} \tilde{\phi}_1 f_{\sigma 0} \int_{-\infty}^t dt' \left\{ \exp[-i\omega t'] \frac{d}{dt'} \exp[i\mathbf{k} \cdot \mathbf{x}(t')] \right\} \\ &= -\frac{q_{\sigma}}{\kappa T_{\sigma}} \tilde{\phi}_1 f_{\sigma 0} \left\{ \left[\exp(i\mathbf{k} \cdot \mathbf{x}(t') - i\omega t') \right]_{-\infty}^t + i\omega I_{\text{phase}}(\mathbf{x}, t) \right\}, \end{aligned} \quad (8.16)$$

where the *phase-history integral* I_{phase} is defined as

$$I_{\text{phase}}(\mathbf{x}, t) = \int_{-\infty}^t dt' \exp(i\mathbf{k} \cdot \mathbf{x}(t') - i\omega t').$$

Evaluation of I_{phase} requires knowledge of the unperturbed orbit trajectory $\mathbf{x}(t')$. This trajectory, determined by solving Eqs. (8.9) with boundary conditions specified by Eq. (8.7), has the velocity time-dependence

$$\mathbf{v}(t') = v_{\parallel} \hat{B} + \mathbf{v}_{\perp} \cos[\omega_{c\sigma}(t' - t)] - \hat{B} \times \mathbf{v}_{\perp} \sin[\omega_{c\sigma}(t' - t)]. \quad (8.17)$$

Equation (8.17) satisfies both the dynamics and the boundary condition $\mathbf{v}(t) = \mathbf{v}$ and so gives the correct helical “unwinding” into the past for a particle at its present position in phase-space. The position trajectory, found by integrating Eq. (8.17), is

$$\begin{aligned} \mathbf{x}(t') &= \mathbf{x} + v_{\parallel} (t' - t) \hat{B} \\ &\quad + \frac{1}{\omega_{c\sigma}} \left\{ \mathbf{v}_{\perp} \sin[\omega_{c\sigma}(t' - t)] + \hat{B} \times \mathbf{v}_{\perp} (\cos[\omega_{c\sigma}(t' - t)] - 1) \right\}, \end{aligned} \quad (8.18)$$

which satisfies the related boundary condition $\mathbf{x}(t) = \mathbf{x}$.

To proceed further, we define φ to be the velocity-space angle between the fixed quantity $\mathbf{k}_{\perp}$ and the dummy variable $\mathbf{v}_{\perp}$ so that $\mathbf{k}_{\perp} \cdot \mathbf{v}_{\perp} = k_{\perp} v_{\perp} \cos \varphi$, and $\mathbf{k}_{\perp} \cdot \hat{B} \times \mathbf{v}_{\perp} = k_{\perp} v_{\perp} \sin \varphi$. Using this definition, the time history of the spatial part of the phase can be written as

$$\mathbf{k} \cdot \mathbf{x}(t') = \mathbf{k} \cdot \mathbf{x} + k_{\parallel} v_{\parallel} (t' - t) + \frac{k_{\perp} v_{\perp}}{\omega_{c\sigma}} \left\{ \sin[\omega_{c\sigma}(t' - t) + \varphi] - \sin \varphi \right\}. \quad (8.19)$$

The phase integral can now be expanded

$$\begin{aligned} I_{\text{phase}}(\mathbf{x}, t) &= e^{-i\omega t} \int_{-\infty}^t dt' e^{i\mathbf{k} \cdot \mathbf{x}(t') - i\omega(t' - t)} \\ &= e^{i\mathbf{k} \cdot \mathbf{x} - i\omega t} \int_{-\infty}^0 d\tau \exp \left\{ i \left[\frac{k_{\perp} v_{\perp}}{\omega_{c\sigma}} \{ \sin[\omega_{c\sigma} \tau + \varphi] - \sin \varphi \} \right] \right\}, \end{aligned} \quad (8.20)$$

where $\tau = t' - t$. Certain Bessel function relations are now of use. The first of these is the integral representation of the J_n Bessel function, namely

$$J_n(z) = \frac{1}{2\pi} \int_0^{2\pi} e^{iz \sin \theta - in\theta} d\theta. \quad (8.21)$$

The inverse of this relation is

$$e^{iz \sin \theta} = \sum_{n=-\infty}^{\infty} J_n(z) e^{in\theta}, \quad (8.22)$$

which may be validated by taking the θ Fourier transform of both sides over the interval from 0 to 2π .

The phase integral can be evaluated using Eq. (8.22) in Eq. (8.20) to obtain

$$\begin{aligned} I_{phase}(\mathbf{x}, t) &= e^{i\mathbf{k} \cdot \mathbf{x} - i\omega t} \sum_{n=-\infty}^{\infty} J_n\left(\frac{k_{\perp} v_{\perp}}{\omega_{c\sigma}}\right) e^{-\frac{ik_{\perp} v_{\perp} \sin \varphi}{\omega_{c\sigma}}} \int_{-\infty}^0 d\tau e^{i(k_{\parallel} v_{\parallel} - \omega)\tau + in(\omega_{c\sigma}\tau + \varphi)} \\ &= e^{i\mathbf{k} \cdot \mathbf{x} - i\omega t} e^{-\frac{ik_{\perp} v_{\perp} \sin \varphi}{\omega_{c\sigma}}} \sum_{n=-\infty}^{\infty} J_n\left(\frac{k_{\perp} v_{\perp}}{\omega_{c\sigma}}\right) \frac{\left[e^{i(k_{\parallel} v_{\parallel} - \omega)\tau + in(\omega_{c\sigma}\tau + \varphi)} \right]_{-\infty}^0}{-i(\omega - k_{\parallel} v_{\parallel} - n\omega_{c\sigma})}. \end{aligned} \quad (8.23)$$

In Eqs. (8.16) and (8.23) there is a lower limit at $t = -\infty$; this limit corresponds to the phase in the distant past and is essentially the information regarding the initial condition of the system in the distant past. We saw in our previous discussion of the Landau problem that initial value problems ought to be analyzed using Laplace transforms, not Fourier transforms. However, if we ignore the initial conditions and use the Plemelj formula with Fourier transforms, the same result as the Laplace method is obtained. This shortcut procedure will now be followed and so any reference to the initial conditions will be dropped and Fourier transforms will be used with invocation of the Plemelj formula whenever it is necessary to resolve any singularities in the integrations. Hence, terms evaluated at $t = -\infty$ in Eqs. (8.16) and (8.23) are dropped since these are initial values. After making these simplifications, the perturbed distribution function becomes

$$f_{\sigma 1}(\mathbf{x}, \mathbf{v}, t) = -\frac{q_{\sigma} \tilde{\phi}_1 f_{\sigma 0} e^{i\mathbf{k} \cdot \mathbf{x} - i\omega t}}{\kappa T_{\sigma}} \left\{ 1 - e^{-\frac{ik_{\perp} v_{\perp} \sin \varphi}{\omega_{c\sigma}}} \sum_n \frac{J_n\left(\frac{k_{\perp} v_{\perp}}{\omega_{c\sigma}}\right) \omega e^{in\varphi}}{(\omega - k_{\parallel} v_{\parallel} - n\omega_{c\sigma})} \right\}. \quad (8.24)$$

The next step is to evaluate the density perturbation Eq. (8.5), an operation that involves integrating the perturbed distribution function over velocity; the velocity integrals are evaluated using the Bessel identity (Watson 1922)

$$\int_0^\infty z J_n^2(\beta z) e^{-\alpha^2 z^2} dz = \frac{1}{2\alpha^2} e^{-\beta^2/2\alpha^2} I_n\left(\frac{\beta^2}{2\alpha^2}\right). \quad (8.25)$$

Using this identity, substitution of Eq. (8.24) into Eq. (8.5) gives

$$\begin{aligned} n_{\sigma 1} &= -\frac{q_\sigma}{\kappa T_\sigma} \phi_1 n_{\sigma 0} \left[\frac{1 - \frac{\omega}{\pi^{3/2} v_{T\sigma}^3} \sum_n \int_{-\infty}^\infty dv_\parallel \int_0^{2\pi} d\varphi \int_0^\infty v_\perp dv_\perp}{e^{-i \frac{k_\perp v_\perp}{\omega_{c\sigma}} \sin \varphi + i n \varphi} J_n\left(\frac{k_\perp v_\perp}{\omega_{c\sigma}}\right) e^{-(v_\parallel^2 + v_\perp^2)/v_{T\sigma}^2}} \right. \\ &\quad \left. \frac{1}{(\omega - k_\parallel v_\parallel - n \omega_{c\sigma})} \right] \\ &= -\frac{q_\sigma \phi_1 n_{\sigma 0}}{\kappa T_\sigma} \left[1 - \frac{2\omega}{\sqrt{\pi} v_{T\sigma}^3} \sum_n \int_{-\infty}^\infty dv_\parallel \int_0^\infty v_\perp dv_\perp \frac{J_n^2\left(\frac{k_\perp v_\perp}{\omega_{c\sigma}}\right) e^{-\frac{v_\parallel^2 + v_\perp^2}{v_{T\sigma}^2}}}{(\omega - k_\parallel v_\parallel - n \omega_{c\sigma})} \right] \\ &= -\frac{q_\sigma}{\kappa T_\sigma} \phi_1 n_{\sigma 0} \left[1 + \alpha_{0\sigma} e^{-k_\perp^2 r_{L\sigma}^2} \sum_n \frac{I_n(k_\perp^2 r_{L\sigma}^2)}{\pi^{1/2}} \int_{-\infty}^\infty d\xi \frac{e^{-\xi^2}}{\xi - \alpha_{n\sigma}} \right] \\ &= -\frac{q_\sigma}{\kappa T_\sigma} \phi_1 n_{\sigma 0} \left[1 + \alpha_{0\sigma} e^{-k_\perp^2 r_{L\sigma}^2} \sum_n I_n(k_\perp^2 r_{L\sigma}^2) Z(\alpha_{n\sigma}) \right], \end{aligned} \quad (8.26)$$

where

$$\alpha_{n\sigma} = \frac{\omega - n \omega_{c\sigma}}{k_\parallel v_{T\sigma}}, \quad (8.27)$$

the Larmor radius is defined to be

$$r_{L\sigma} = \frac{\sqrt{\kappa T_\sigma / m_\sigma}}{\omega_{c\sigma}}, \quad (8.28)$$

and Z is the plasma dispersion function defined in Eq. (5.70). Finally, Eq. (8.26) is substituted into Eq. (8.4) to obtain the warm magnetized plasma electrostatic dispersion relation

$$\mathcal{D}(\omega, k) = 1 + \sum_\sigma \frac{1}{k^2 \lambda_{D\sigma}^2} \left[1 + \alpha_{0\sigma} e^{-k_\perp^2 r_{L\sigma}^2} \sum_{n=-\infty}^\infty I_n(k_\perp^2 r_{L\sigma}^2) Z(\alpha_{n\sigma}) \right] = 0. \quad (8.29)$$

Note that $\mathcal{D}(\omega, k)$ refers to the dispersion relation and should not be confused with D , the off-diagonal term of the cold plasma dielectric tensor, nor with $\mathbf{D}$, the displacement vector. A similar, but more involved calculation using unperturbed

orbit phase integrals for the perturbed current density gives the hot plasma version of the full electromagnetic dispersion, i.e., the finite temperature generalization of the cold plasma dielectric tensor $\overleftrightarrow{\mathbf{K}}$. Although the calculation is essentially similar, it is considerably more tedious, and the interested reader is referred to specialized texts on plasma waves such as those by Stix (1992) or Swanson (2003).

8.2 Analysis of the warm plasma electrostatic dispersion relation

Equation (8.29) generalizes the unmagnetized warm plasma electrostatic dispersion relation Eq. (5.55) to the situation of a magnetized warm plasma. Thus, Eq. (8.29) ought to revert to Eq. (5.55) in the $B \rightarrow 0$ limit. The Bessel identity

$$\sum_{n=-\infty}^{\infty} I_n(\lambda) = e^{\lambda} \quad (8.30)$$

together with the condition $\alpha_n \rightarrow \alpha_0$ if $B \rightarrow 0$ show that this is indeed the case.

Furthermore, Eq. (8.29) should also be the warm plasma generalization of the cold, magnetized plasma, electrostatic dispersion

$$k_x^2 S + k_z^2 P = 0; \quad (8.31)$$

demonstration of this correspondence will be presented later. Equation (8.30) can be used to recast Eq. (8.29) as

$$\mathcal{D}(\omega, k) = 1 + \sum_{\sigma} \frac{e^{-k_{\perp}^2 r_{L\sigma}^2}}{k^2 \lambda_{D\sigma}^2} \sum_{n=-\infty}^{\infty} I_n(k_{\perp}^2 r_{L\sigma}^2) [1 + \alpha_{0\sigma} Z(\alpha_{n\sigma})] = 0. \quad (8.32)$$

Using $I_{-n}(z) = I_n(z)$, the summation over n can be rearranged to give

$$\mathcal{D}(\omega, k) = 1 + \sum_{\sigma} \frac{e^{-k_{\perp}^2 r_{L\sigma}^2}}{k^2 \lambda_{D\sigma}^2} \left\{ \frac{I_0(k_{\perp}^2 r_{L\sigma}^2) (1 + \alpha_{0\sigma} Z(\alpha_{0\sigma})) + \sum_{n=1}^{\infty} I_n(k_{\perp}^2 r_{L\sigma}^2) [2 + \alpha_{0\sigma} \{Z(\alpha_{n\sigma}) + Z(\alpha_{-n\sigma})\}]}{\sum_{n=1}^{\infty} I_n(k_{\perp}^2 r_{L\sigma}^2) [2 + \alpha_{0\sigma} \{Z(\alpha_{n\sigma}) + Z(\alpha_{-n\sigma})\}]} \right\}. \quad (8.33)$$

In order to obtain the lowest order thermal correction, it is assumed $\alpha_0 = \omega/k_{\parallel} v_{T\sigma} \gg 1$, in which case the large argument expansion Eq. (5.77) can be used to evaluate $Z(\alpha_0)$. Invoking this expansion, and keeping only lowest order terms, shows that

$$\begin{aligned} 1 + \alpha_0 Z(\alpha_0) &= 1 + \alpha_0 \left\{ -\frac{1}{\alpha_0} \left[1 + \frac{1}{2\alpha_0^2} + \dots \right] + i\pi^{1/2} \exp(-\alpha_0^2) \right\} \\ &= -\frac{1}{2\alpha_0^2} + i\alpha_0 \pi^{1/2} \exp(-\alpha_0^2). \end{aligned} \quad (8.34)$$

It is additionally assumed that $|\alpha_n| = |(\omega - n\omega_{c\sigma})/k_{\parallel}v_{T\sigma}| \gg 1$ for $n \neq 0$; this corresponds to assuming that the wave frequency is not too close to a cyclotron resonance. Because of this assumption, the large argument expansion of the plasma dispersion function is also appropriate for the $n \neq 0$ terms, and so we can write

$$\begin{aligned} 2 + \alpha_0 [Z(\alpha_n) + Z(\alpha_{-n})] &= 2 + \alpha_0 \left[-\frac{1}{\alpha_n} - \frac{1}{\alpha_{-n}} + i\sqrt{\pi} (e^{-\alpha_n^2} + e^{-\alpha_{-n}^2}) \right] \\ &= 2 - \frac{\omega}{(\omega - n\omega_{c\sigma})} - \frac{\omega}{(\omega + n\omega_{c\sigma})} + i\alpha_0\sqrt{\pi} (e^{-\alpha_n^2} + e^{-\alpha_{-n}^2}) \\ &= -\frac{2n^2\omega_{c\sigma}^2}{(\omega^2 - n^2\omega_{c\sigma}^2)} + i\alpha_0\sqrt{\pi} (e^{-\alpha_n^2} + e^{-\alpha_{-n}^2}), \end{aligned} \quad (8.35)$$

where the subscript σ has been omitted from the α_n to keep the algebra uncluttered. Substitution of these expansions back into the dispersion relation gives

$$\begin{aligned} \mathcal{D}(\omega, k) &= 1 + \sum_{\sigma} \frac{e^{-k_{\perp}^2 r_{L\sigma}^2}}{k^2 \lambda_{D\sigma}^2} \left\{ I_0(k_{\perp}^2 r_{L\sigma}^2) \left(-\frac{1}{2\alpha_{0\sigma}^2} + i\alpha_{0\sigma}\sqrt{\pi} \exp(-\alpha_{0\sigma}^2) \right) + \right. \\ &\quad \left. \sum_{n=1}^{\infty} I_n(k_{\perp}^2 r_{L\sigma}^2) \begin{pmatrix} -\frac{2n^2\omega_{c\sigma}^2}{\omega^2 - n^2\omega_{c\sigma}^2} \\ +i\alpha_{0\sigma}\sqrt{\pi} \begin{pmatrix} e^{-\alpha_n^2} \\ +e^{-\alpha_{-n}^2} \end{pmatrix} \end{pmatrix} \right\} \\ &= 0. \end{aligned} \quad (8.36)$$

Equation (8.36) generalizes the magnetized cold plasma electrostatic dispersion to finite temperature and shows how Landau damping appears both at the wave frequency ω and also at cyclotron harmonics, i.e., in the vicinity of $n\omega_{c\sigma}$. Two important limits of Eq. (8.36) are discussed in the following sections.

8.3 Bernstein waves

Suppose the wave phase is uniform in the direction along the magnetic field. Such a situation would occur if the antenna exciting the wave were an infinitely long wire aligned parallel to a magnetic field line (in reality, the antenna would have to be sufficiently long to behave as if infinite). This situation corresponds

to having $k_{\parallel} \rightarrow 0$, in which case the Landau damping terms and the $1/2\alpha_{0\sigma}^2$ term vanish. The dispersion relation Eq. (8.36) consequently reduces to

$$1 = \sum_{\sigma} \frac{e^{-\lambda_{\sigma}}}{\lambda_{\sigma}} \sum_{n=1}^{\infty} \frac{2n^2 \omega_{p\sigma}^2}{\omega^2 - n^2 \omega_{c\sigma}^2} I_n(\lambda_{\sigma}), \quad (8.37)$$

where

$$\lambda_{\sigma} = k_{\perp}^2 r_{L\sigma}^2 \quad (8.38)$$

and

$$r_{L\sigma}^2 = \frac{\omega_{p\sigma}^2 \lambda_{D\sigma}^2}{\omega_{c\sigma}^2} = \frac{\kappa T_{\sigma}}{m_{\sigma} \omega_{c\sigma}^2} \quad (8.39)$$

is the Larmor radius. The waves resulting from this dispersion were first derived by Bernstein (1958) and are called Bernstein or cyclotron harmonic waves; they are also sometimes referred to as hot plasma waves because their existence depends on the plasma having a finite temperature. Bernstein waves have been observed in both laboratory experiments and in spacecraft measurements of magnetospheric plasmas. Early measurements of electron Bernstein waves in laboratory experiments were reported by Crawford (1965) and by Leuterer (1969); measurements of ion Bernstein waves were reported by Schmitt (1973). An example of a spacecraft measurement was reported by Moncuquet, Meyervernet, and Hoang (1995) who used a fit of data to the Bernstein dispersion relation to infer the electron temperature in the magnetized plasma of Jupiter's moon, Io. Electron Bernstein waves involve ω being in the vicinity of an electron cyclotron harmonic, i.e., $\omega^2 \sim \mathcal{O}(n^2 \omega_{ce}^2)$, whereas for ion Bernstein waves, ω is in the vicinity of an ion cyclotron harmonic.

The Bernstein wave dispersion relation, Eq. (8.37), provides a transcendental relation between ω and $k_{\perp}$. For any given $k_{\perp}$ the dispersion relation has an infinite number of roots ω , each associated with a different cyclotron harmonic. This is because as ω is increased from small values to infinity, the $(\omega^2 - n^2 \omega_{c\sigma}^2)^{-1}$ term on the right-hand side of Eq. (8.37) will cause the entire right-hand side to oscillate between minus and plus infinity at each successive cyclotron harmonic. In particular, the right-hand side will assume a value of $+\infty$ when ω^2 is infinitesimally larger than $n^2 \omega_{c\sigma}^2$ and a value of $-\infty$ when ω^2 is infinitesimally smaller than $n^2 \omega_{c\sigma}^2$. Thus, as ω increases from the n th cyclotron harmonic to the $(n+1)$ th cyclotron harmonic, the right-hand side of Eq. (8.37) will take on all values between minus and plus infinity and so will always equal the left-hand side at some value of ω in this range.

We now consider electron Bernstein waves and so drop the ion terms from the summation (the analysis for ion Bernstein waves is similar). The subscript σ

will be dropped and it will be understood that all quantities refer to electrons. On keeping the electron terms only, Eq. (8.37) reduces to

$$1 = 2 \frac{e^{-\lambda}}{\lambda} \sum_{n=1}^{\infty} \frac{n^2 \omega_p^2}{\omega^2 - n^2 \omega_c^2} I_n(\lambda), \quad (8.40)$$

which has the following limiting forms depending on the ratio of ω_p^2 to ω_c^2 :

1. $\omega_p^2 \ll \omega_c^2$ case:

Here each $2n^2 \omega_p^2 / (\omega^2 - n^2 \omega_c^2)$ factor is negligible compared to unity except when $\omega^2 \sim \mathcal{O}(n^2 \omega_c^2)$. Thus, for a given ω only one term in the summation is near resonance and the dispersion relation is satisfied by this one term on the right-hand side of Eq. (8.40) balancing the left-hand side. This results in an infinite set of modes, each slightly above a cyclotron harmonic, i.e.,

$$\omega^2 = n^2 \omega_c^2 + 2 \frac{e^{-\lambda}}{\lambda} n^2 \omega_p^2 I_n(\lambda), \quad \text{for } n = 1, 2, \dots, \infty. \quad (8.41)$$

The small and large λ limits of this expression are determined using the asymptotic values of the I_n Bessel function, namely for $n \geq 1$

$$\begin{aligned} \lim_{\lambda \ll 1} I_n(\lambda) &= \frac{1}{n!} \left(\frac{\lambda}{2} \right)^n \\ \lim_{\lambda \gg 1} I_n(\lambda) &= \frac{1}{\sqrt{2\pi\lambda}} e^{\lambda} \end{aligned} \quad (8.42)$$

to give

$$\begin{aligned} \omega^2 &= n^2 \omega_c^2 + (1 - \lambda) n^2 \omega_p^2 \frac{1}{n!} \left(\frac{\lambda}{2} \right)^{n-1} \quad \text{for } \lambda \ll 1, \\ \omega^2 &= n^2 \omega_c^2 \quad \text{for } \lambda \gg 1. \end{aligned} \quad (8.43)$$

The $n = 1$ mode differs slightly from the $n \geq 2$ modes because the $n = 1$ mode has the small λ dispersion

$$\omega^2 = \omega_{uh}^2 - \lambda \omega_p^2 \quad (8.44)$$

showing this mode is the warm plasma generalization of the upper hybrid resonance. The frequency of the $n \geq 2$ modes starts at $\omega^2 = n^2 \omega_c^2$ when $\lambda = 0$, takes on a maximum value at some finite λ , and then reverts to $\omega^2 = n^2 \omega_c^2$ as $\lambda \rightarrow \infty$. To the left of the frequency maximum the group velocity $\partial\omega/\partial k_{\perp}$ is positive and to the right of the maximum the group velocity is negative. This system of modes is sketched in Fig. 8.1(a).

2. $\omega_p^2 \gg \omega_c^2$ case:

For large λ , the product of the exponential factor and the modified Bessel function in Eq. (8.40) is unity, leaving a $\lambda^{3/2}$ factor in the denominator, so again the dispersion reduces to cyclotron harmonics, $\omega^2 = n^2 \omega_c^2$ as $\lambda \rightarrow \infty$. For small λ , the situation is

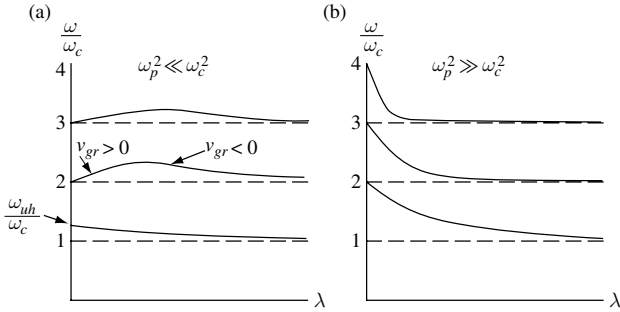


Fig. 8.1 Electron Bernstein wave dispersion relations.

more involved, because the $n = 1$ term is independent of λ to lowest order and so the $n = 1$ term must always be retained when approximating Eq. (8.40). Let us suppose some other term, say the $(n + 1)$ th term, is near resonance, i.e., $\omega^2 \sim (n + 1)^2 \omega_c^2$. Then, keeping the left-hand side, the $n = 1$ term, and the $(n + 1)$ th term, expansion of Eq. (8.40) results in

$$1 \simeq \frac{\omega_p^2}{[(n + 1)^2 - 1] \omega_c^2} + \frac{(n + 1) \omega_p^2}{[\omega^2 - (n + 1)^2 \omega_c^2] n!} \left(\frac{\lambda}{2}\right)^n, \tag{8.45}$$

which may be solved for ω to give

$$\omega = (n + 1) \omega_c \left\{ 1 - \frac{\omega_p^2 [(n + 1)^2 - 1]}{(\omega_p^2 - [(n + 1)^2 - 1] \omega_c^2)} \frac{\lambda^n}{2^{n+1} (n + 1)!} \right\}. \tag{8.46}$$

Thus, the n th mode starts at $\omega \simeq (n + 1) \omega_c$ for small λ and then the frequency decreases towards the asymptotic limit $n \omega_c$ as $\lambda \rightarrow \infty$. This dispersion is sketched in Fig. 8.1(b).

Excitation of Bernstein waves requires setting $k_{\parallel} = 0$, which turns out to be a quite stringent requirement. The antenna must be absolutely uniform in the direction along the magnetic field because otherwise a finite $k_{\parallel}$ will be excited, which would result in finite instead of infinite $\omega/k_{\parallel}$ and so cause the waves to be subject to strong Landau damping.

8.4 Finite $k_{\parallel}$ dispersion: linear mode conversion

An alternate limit for Eq. (8.36) would be to allow $k_{\parallel}$ to be finite but also have $k_{\perp}^2 r_L^2 \ll 1$ so only the lowest-order finite Larmor radius terms are retained. To keep matters simple, and also to relate to cold plasma theory, $k_{\parallel}$ will be assumed to be

sufficiently small to make the wave-phase velocity much faster than the particle thermal velocities, i.e., $\omega/k_{\parallel} \gg v_{Te}, v_{Ti}$. Since $r_{L\sigma}^2/\lambda_{D\sigma}^2 = \omega_{p\sigma}^2/\omega_{c\sigma}^2$, the lowest-order perpendicular thermal terms will be $\mathcal{O}(k_{\perp}^4)$ and so perpendicular quantities up to fourth order must be retained. This means that both the $n = 1$ and the $n = 2$ terms must be retained in the summation over n . With these approximations, and using $1/2\lambda_{D\sigma}^2\alpha_0^2 = \omega_p^2/\omega^2$, Eq. (8.36) becomes

$$k_{\parallel}^2 + k_{\perp}^2 + \sum_{\sigma} (1 - k_{\perp}^2 r_{L\sigma}^2) \left(\left(1 + \frac{k_{\perp}^4 r_{L\sigma}^4}{4} \right) \left(-\frac{\omega_{p\sigma}^2}{\omega^2} + i \frac{\alpha_0 \sqrt{\pi}}{\lambda_{D\sigma}^2} e^{-\alpha_0^2} \right) + \frac{k_{\perp}^2 \omega_{p\sigma}^2}{2\omega_{c\sigma}^2} \left[\frac{-2\omega_{c\sigma}^2}{(\omega^2 - \omega_{c\sigma}^2)} + i\alpha_0 \sqrt{\pi} \left(\frac{e^{-\alpha_1^2}}{+e^{-\alpha_{-1}^2}} \right) \right] + \frac{k_{\perp}^4 r_{L\sigma}^2 \omega_{p\sigma}^2}{8\omega_{c\sigma}^2} \left[\frac{-8\omega_{c\sigma}^2}{(\omega^2 - 4\omega_{c\sigma}^2)} + i\alpha_0 \sqrt{\pi} \left(\frac{e^{-\alpha_2^2}}{+e^{-\alpha_{-2}^2}} \right) \right] \right) = 0. \quad (8.47)$$

The Landau damping terms will be assumed to be negligible to keep matters simple and the equation will now be grouped according to powers of $k_{\perp}^2$. Retaining only the lowest-order, finite-temperature perpendicular terms gives

$$k_{\parallel}^2 \left(1 - \sum_{\sigma} \frac{\omega_{p\sigma}^2}{\omega^2} \right) + k_{\perp}^2 \left(1 - \sum_{\sigma} \frac{\omega_{p\sigma}^2}{(\omega^2 - \omega_{c\sigma}^2)} \right) - k_{\perp}^4 \sum_{\sigma} \left(\frac{3\omega_{p\sigma}^4 \lambda_{D\sigma}^2}{(\omega^2 - \omega_{c\sigma}^2)(\omega^2 - 4\omega_{c\sigma}^2)} \right) = 0. \quad (8.48)$$

This is of the form

$$-k_{\perp}^4 \epsilon_{th} + k_{\perp}^2 S + k_{\parallel}^2 P = 0, \quad (8.49)$$

where the perpendicular fourth-order thermal coefficient is

$$\epsilon_{th} = \sum_{\sigma} \left(\frac{3\omega_{p\sigma}^4 \lambda_{D\sigma}^2}{(\omega^2 - \omega_{c\sigma}^2)(\omega^2 - 4\omega_{c\sigma}^2)} \right). \quad (8.50)$$

Equation (8.49) is a quadratic equation in $k_{\perp}^2$. The cold plasma model used earlier in effect set $\epsilon_{th} = 0$, in which case a wave propagating through an inhomogeneous plasma towards an $S = 0$ hybrid resonance would have $k_{\perp}^2 \rightarrow \infty$. This non-physical prediction is resolved by the warm plasma theory because ϵ_{th} , while

small, is finite and prevents $k_{\perp}^2 \rightarrow \infty$ from occurring. What happens instead is that Eq. (8.49) has two qualitatively distinct roots, namely

$$k_{\perp}^2 = \frac{S \pm \sqrt{S^2 + 4\epsilon_{th}k_{\parallel}^2P}}{2\epsilon_{th}}. \quad (8.51)$$

At locations far from the hybrid resonance, S is large, in which case the two modes are well separated and given by

$$\begin{aligned} k_{\perp}^2 &= -\frac{P}{S}k_{\parallel}^2, & \text{cold plasma wave,} \\ k_{\perp}^2 &= \frac{S}{\epsilon_{th}}, & \text{hot plasma wave,} \end{aligned} \quad (8.52)$$

but at locations near the hybrid resonance S approaches zero and the two modes coalesce. The hot plasma wave results from balancing the first and middle terms in Eq. (8.49) while the cold plasma wave results from balancing the middle and last terms. The actual mode coalescence occurs where the square root term in Eq. (8.51) vanishes, i.e., where

$$S^2 = -4\epsilon_{th}k_{\parallel}^2P. \quad (8.53)$$

At the location of mode coalescence $dk_{\perp}/dx$ is infinite, violating the $dk_{\perp}/dx \ll k_x^2$ requirement for the WKB approximation to be valid. Thus, a more sophisticated analysis than WKB is required. Such an analysis is presented in the next section where hot and cold waves are shown to become strongly coupled in the coalescence region. This strong coupling between the two modes should not be surprising because the two modes are essentially indistinguishable at this location. It will be demonstrated in particular that a cold wave propagating towards the hybrid resonance (e.g., see Fig. 8.2) can be linearly converted into a hot wave, which then propagates back outwards from the resonance.

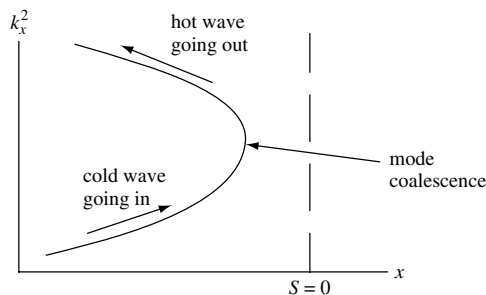


Fig. 8.2 Linear mode conversion of cold mode into hot mode near hybrid resonance.

8.5 Analysis of linear mode conversion

8.5.1 Airy equation

The procedure for analyzing linear mode conversion was developed by Stix (1965) and is an extension of the standard method for connecting solutions of the Airy equation

$$y'' + xy = 0 \quad (8.54)$$

from the $x < 0$ region to the $x > 0$ region. The Airy problem involves a second-order ordinary differential equation with a term containing a factor that vanishes at some location, namely the term xy vanishes at $x = 0$. The mode conversion problem involves a fourth-order equation, which also contains a term with a factor that vanishes at some location, namely S vanishes at a hybrid resonance. The Airy connection method will first be examined in order to introduce the relevant concepts and then these concepts will be applied to the actual mode conversion problem.

The sequence of steps for developing the solution to the Airy problem are as follows:

1. Laplace transform the equation. Equation (8.54) cannot be Fourier transformed because of the non-constant coefficient x in the second term; however, it can be Laplace transformed by taking advantage of the relation that the Laplace transform of $xy(x)$ is $-d\tilde{y}(p)/dp$. Thus, the Laplace transform of Eq. (8.54) is

$$p^2\tilde{y}(p) - \frac{d\tilde{y}(p)}{dp} = 0. \quad (8.55)$$

Strictly speaking, Laplace transforming y'' will also introduce terms involving initial conditions, but these additional terms will be ignored for now and this issue will be addressed later. Equation (8.55) is easily solved to give

$$\tilde{y}(p) = A \exp(p^3/3), \quad (8.56)$$

where A is a constant. The formal solution to Eq. (8.54) is the inverse transform

$$y(x) = (2\pi i)^{-1} \int_C \tilde{y}(p) e^{px} dp = a \int_C e^{f(p)} dp, \quad (8.57)$$

where $a = A/2\pi i$, $f(p) = p^3/3 + px$, and C is some contour in the complex plane. By appropriate choice of this contour Eq. (8.57) is shown to be a valid solution to the original differential equation. In particular, if the endpoints p_1 and p_2 of the contour C are chosen to satisfy

$$[e^{f(p)}]_{p_1}^{p_2} = 0, \quad (8.58)$$

then Eq. (8.57) turns out to be a valid solution to Eq. (8.54). The reason for this may be seen by explicitly calculating y'' and xy using Eq. (8.57) to obtain

$$y'' = \frac{d^2}{dx^2} a \int_C e^{p^3/3+px} dp = a \int_C p^2 e^{p^3/3+px} dp \quad (8.59)$$

and

$$\begin{aligned} xy &= a \int_C e^{p^3/3} \left(\frac{d}{dp} e^{px} \right) dp \\ &= a \left[e^{p^3/3+px} \right]_{p_1}^{p_2} - a \int_C \frac{d}{dp} (e^{p^3/3}) e^{px} dp \\ &= -a \int_C p^2 e^{p^3/3+px} dp. \end{aligned} \quad (8.60)$$

Addition of these results corresponds to Eq. (8.54) if the end-point boundary conditions specified by Eq. (8.58) are satisfied.

2. Suppose the integrand is *analytic* in some region and consider two different contours in this region having the *same* pair of endpoints. We can take one of these contours and deform it into the other. Thus, the two contours give the same result and so correspond to the same solution.
3. Independence of contours. Since Eq. (8.54) is a second-order ordinary differential equation it must have two linearly independent solutions. This means there must be two linearly independent contours since a choice of contour corresponds to a solution. Linear independence of the contours means one contour cannot be deformed into the other. One possible way for the two contours to be linearly independent is for them to have different pairs of endpoints. In this case, one contour can only be deformed into the other if the endpoints can be moved without changing the integral. If moving the endpoints changes the integral, then the two contours are independent. Another possibility is the situation where the two contours have the same pairs of endpoints, but the integrand is *not analytic* in the region between the two contours. For example, there could be a pole or branch cut between the two contours. Then, one contour could not be deformed into the other because of the non-analytic region separating the two contours. The two contours would then be linearly independent.

8.5.2 Steepest descent contour

The solution to the Airy equation is thus

$$y(x) = \int_C e^{f(p,x)} dp, \quad (8.61)$$

where C satisfies the conditions listed above and the solution can be multiplied by an arbitrary constant. In order to evaluate the integral, it is useful to first separate the complex function $f(p, x)$ into its real and imaginary parts,

$$f(p, x) = f_r(p, x) + i f_i(p, x). \quad (8.62)$$

Also, it should be remembered that, for the purposes of the p integration, x can be considered as a fixed parameter. Thus, when performing the p integration, x need not be written explicitly and so we may simply write $f(p) = f_r(p) + i f_i(p)$, where $p = u + iv$ and u, v are the coordinates in the complex p -plane. For an arbitrary contour in the p -plane, both f_r and f_i will vary. The variation of the phase factor f_i means there will be alternating positive and negative contributions to the integral in Eq. (8.61), making evaluation of this integral very delicate. However, evaluation of the integral can be made almost trivial if the contour is deformed to follow a certain optimum path.

To see this, it should first be recalled from the theory of complex variables that any function of a complex variable must satisfy the Cauchy–Riemann conditions; this means f_r and f_i must satisfy

$$\frac{\partial f_r}{\partial u} = \frac{\partial f_i}{\partial v}, \quad \frac{\partial f_r}{\partial v} = -\frac{\partial f_i}{\partial u}. \quad (8.63)$$

Also, on defining the $u - v$ plane gradient operator $\nabla = \hat{u} \partial / \partial u + \hat{v} \partial / \partial v$ it is seen that

$$\nabla f_r \cdot \nabla f_i = \frac{\partial f_r}{\partial u} \frac{\partial f_i}{\partial u} + \frac{\partial f_r}{\partial v} \frac{\partial f_i}{\partial v} = 0. \quad (8.64)$$

This means that contours of constant f_r are everywhere orthogonal to contours of constant f_i .

Since f_i corresponds to the phase of $\exp(f_r(p) + i f_i(p))$ and f_r corresponds to the amplitude, Eq. (8.64) shows that a path in the complex p -plane, which is arranged to follow the gradient of f_r , will automatically be a contour of constant f_i , i.e., a contour of constant phase. Thus, the path that is optimum for purposes of evaluation is the path obtained by following ∇f_r because on this path f_i will be constant and so there will not be any alternating positive and negative oscillating contributions to the integral resulting from variation in f_i . In fact, because f_i is constant on this special path, it can be factored from the integral, giving

$$y(x) = e^{i f_i} \int_{C \parallel \nabla f_r} e^{f_r(p)} dp. \quad (8.65)$$

Clearly, the maximum contribution to this integral comes from the vicinity of where f_r assumes its maximum value. Since this maximum occurs where $\nabla f_r = 0$, most of the contribution to the integral comes from the vicinity of where $\nabla f_r = 0$. The extrema of f are always saddle points because the Cauchy–Reimann conditions imply $\partial^2 f_{r,i} / \partial u^2 + \partial^2 f_{r,i} / \partial v^2 = 0$. Thus, the vicinity of $\nabla f_r = 0$ must be a saddle point.

The discussion in the previous paragraph implies that once the endpoints of the contour have been chosen, for purposes of evaluation it is advantageous to deform the contour to follow the gradient of f_r ; this is called the steepest ascent/descent

path (usually just called steepest descent). This optimum choice of path ensures that (i) there is a localized region where f_r assumes a maximum value and (ii) the phase does not vary along the path. The main contribution to the integral is concentrated into a small region of the complex p -plane in the vicinity of where f_r has its maximum value. Simple integration techniques may be used to evaluate the integral in this vicinity, and the contribution from this region dominates other contributions because $\exp(f_r(p))$ is exponentially larger at the maximum of f_r than at other locations. Furthermore, if the contour is chosen so that $f_r \rightarrow -\infty$ at the endpoints then Eq. (8.58) will be satisfied and the chosen contour will be a solution of the original differential equation.

The saddle points are located where $f'(p) = 0$. For the Airy function, the function $f(p)$ and its first and second derivatives are

$$\begin{aligned} f(p) &= p^3/3 + px \\ f'(p) &= p^2 + x \\ f''(p) &= 2p, \end{aligned} \quad (8.66)$$

so the saddle points, found by setting $f'(p) = 0$ and denoted by the subscript s , are located at $p_s = \pm(-x)^{1/2}$. In the vicinity of a saddle point, f can be Taylor expanded as

$$f(p) \simeq f(p_s) + \frac{(p - p_s)^2}{2} f''(p_s). \quad (8.67)$$

It is now convenient to define the origin of the coordinate system to be at the saddle point and also use cylindrical coordinates, r, θ , so $p - p_s = r \exp(i\theta)$ and $dp = dr e^{i\theta}$ for fixed θ . Also, phasor notation is used for f'' so $f'' = |f''| \exp(i\psi)$, where ψ is the phase of f'' . Thus, in the vicinity of a saddle point we write

$$f(p) \simeq f(p_s) + r^2 \frac{|f''(p_s)|}{2} \exp(2i\theta + i\psi). \quad (8.68)$$

Choosing the contour to follow the path of steepest descent corresponds to choosing θ such that $2\theta + \psi = \pm\pi$, in which case

$$f(p) \simeq f(p_s) - r^2 \frac{|f''(p_s)|}{2} \quad (8.69)$$

and so Eq. (8.61) becomes

$$y(x) \simeq \int_{-\infty}^{\infty} e^{f(p_s) - \frac{1}{2} r^2 |f''(p_s)|} dr e^{i\theta}, \quad (8.70)$$

where the approximation has been made that nearly all the contribution to the integral comes from small r . Inaccuracy of the Taylor expansion at large r is of no consequence, since the contributions to the integral from large r are negligible

because of the exponentially decaying behavior of the integrand at large r . Thus, f_r is maximum at the saddle point $r = 0$ and the main contribution to the integral comes from the contour going over the ridge of the saddle. The r integral is a Gaussian integral $\int \exp(-ar^2)dr = \sqrt{\pi/a}$ so

$$\begin{aligned}
 y(x) &\simeq \int_{-\infty}^{\infty} e^{f(p_s) - \frac{1}{2}r^2|f''(p_s)|} dr e^{i\theta} \\
 &= e^{f(p_s) + i\theta} \int_{-\infty}^{\infty} e^{-\frac{1}{2}r^2|f''(p_s)|} dr \\
 &= e^{f(p_s) + i\theta} \sqrt{\frac{2\pi}{|f''(p_s)|}} \\
 &= e^{f(p_s)} \sqrt{\frac{2\pi e^{i2\theta}}{|f''(p_s)|}} \\
 &= e^{f(p_s)} \sqrt{\frac{\pi e^{i(\pm\pi - \psi)}}{|f''(p_s)|}} \\
 &= e^{f(p_s)} \sqrt{-\frac{2\pi}{f''(p_s)}}. \tag{8.71}
 \end{aligned}$$

Since $f(p_s) = p(p^2/3 + x) = \pm\sqrt{-x}(2x/3) = \mp 2(-x)^{3/2}/3$ and $f''(p_s) = \pm 2\sqrt{-x}$, it is seen that

$$\int_{\text{vicinity of saddle}} e^{f(p)} dp \simeq \sqrt{\frac{\mp\pi}{(-x)^{1/2}}} e^{\mp 2(-x)^{3/2}/3}. \tag{8.72}$$

Providing x is not too close to zero, the two saddle points determined by the $\pm$ signs are well separated and do not perturb each other; the integral will then be a summation over whichever saddle points the contour happens to pass over. The polarity of a saddle point is determined by the sense in which the contour passes through the saddle point.

8.5.3 Relationship between saddle-point solutions and WKB modes

When $|x|$ is large, Eq. (8.54) can be solved approximately using the WKB method, in which case it is assumed that $y(x) = A(x) \exp(i \int^x k(x') dx')$. The wavenumber $k(x)$ is determined from the dispersion relation associated with the original differential equation and $A(x)$ is a function of $k(x)$. At large $|x|$ the dispersion relation associated with Eq. (8.54) is obtained by assuming $d/dx \rightarrow ik$ and has the form

$$k^2 = x \tag{8.73}$$

so $k(x) = \pm x^{1/2}$ and $A(x) \sim (\pm x)^{-1/4}$. The WKB solution is thus

$$y_{WKB} \sim \frac{1}{(\pm x)^{1/4}} e^{\pm i \int^x x'^{1/2} dx'} = \frac{1}{(\pm x)^{1/4}} e^{\pm i 2x^{3/2}/3}, \quad (8.74)$$

which is the same as the saddle-point solution. Thus, *each saddle-point solution corresponds to a WKB mode*, and in particular to a propagating wave if $x > 0$ and to an evanescent mode if $x < 0$.

8.5.4 Using boundary conditions to choose contours

The choice of contour is determined by the boundary conditions, which in turn are determined by the physical considerations. In this Airy equation problem, the boundary condition would typically be the physical constraint that no exponentially growing solutions are allowed for $x < 0$. This forces choice of a contour that does *not* pass through the saddle point having the plus sign in Eq. (8.74) when $x < 0$. This condition together with Eq. (8.58) uniquely determines the contour. Choosing a contour C is equivalent to specifying a particular solution to Eq. (8.57); this chosen solution can then be evaluated for $x > 0$. For $x > 0$ the saddle points are at different locations and so the steepest descent path (i.e., the path where evaluation is easiest) will be different, since it will have to pass through the saddle points as they exist for $x > 0$. The sum of the contributions of these saddle points gives the form of the solution for $x > 0$. The important point to realize here is that the *same* contour is used for both $x > 0$ and $x < 0$ because the linearly independent solution is determined by the contour. The contour is determined by its endpoints, but may be deformed provided analyticity is preserved. Thus, the endpoints are the same for both the $x > 0$ and $x < 0$ evaluations, but the deformations of the contour differ in order to pass through the respective $x > 0$ or $x < 0$ saddle points. The sign of x determines the character of the saddle points and so *for purposes of evaluation* the contour is deformed differently for $x > 0$ and $x < 0$.

This completes the discussion of the Airy problem and we now return to the linear mode conversion problem (Stix 1965). A comparison of Eqs. (6.108) and (8.49) shows that if Eq. (8.49) is “un-Fourier” analyzed in the x direction, it becomes

$$\frac{d}{dx} \left(\epsilon_{th} \frac{d^3}{dx^3} + S \frac{d}{dx} \right) \phi - k_{\parallel}^2 P \phi = 0, \quad (8.75)$$

where it is assumed that ϵ_{th} is approximately constant. To prove this is the correct form, consider the following two statements: (i) if $\epsilon_{th} \rightarrow 0$, Eq. (8.75) reverts to Eq. (6.108) (Fourier-analyzed in the z direction, but not the x direction), (ii) if

ϵ_{th} is finite and S and P are assumed uniform, then Fourier analysis in x restores Eq. (8.49).

We now suppose the plasma is non-uniform in a manner such that $S(x) = 0$ at some particular value of x . The x origin is defined to be at this location and so corresponds to the location of the hybrid resonance. In order to be specific, we assume that we are dealing with a mode where $S > 0$ and $P < 0$, since S and P must have opposite signs for the cold plasma wave to propagate. Then, in the vicinity of $x = 0$, Taylor expansion of S gives $S = S'x$ so that Eq. (8.75) becomes

$$\epsilon_{th} \frac{d^4 \phi}{dx^4} + S' \frac{d\phi}{dx} + S'x \frac{d^2 \phi}{dx^2} - k_{\parallel}^2 P \phi = 0, \quad (8.76)$$

which has a coefficient that vanishes at $x = 0$ just like the Airy equation. Although Eq. (8.76) could be analyzed as it stands, it is better to tidy its format by changing to a suitably chosen dimensionless coordinate. This is done by first defining

$$\xi = \frac{x}{\lambda}, \quad (8.77)$$

where λ is an as-yet undetermined characteristic length that will be chosen to provide maximum simplification of the coefficients. Replacing x by ξ in Eq. (8.76) gives

$$\frac{\epsilon_{th}}{\lambda^4} \frac{d^4 \phi}{d\xi^4} + \frac{S'}{\lambda} \frac{d\phi}{d\xi} + \frac{S'}{\lambda} \xi \frac{d^2 \phi}{d\xi^2} - k_{\parallel}^2 P \phi = 0. \quad (8.78)$$

This becomes

$$\frac{d^4 \phi}{d\xi^4} + \frac{d\phi}{d\xi} + \xi \frac{d^2 \phi}{d\xi^2} + \mu \phi = 0 \quad (8.79)$$

by choosing the scaling constants λ and μ to satisfy $\epsilon_{th}/S'\lambda^3 = 1$ and $\mu = -k_{\parallel}^2 \lambda P/S'$. These choices imply

$$\lambda = \left(\frac{\epsilon_{th}}{S'} \right)^{1/3}, \quad \mu = -\frac{k_{\parallel}^2 P \epsilon_{th}^{1/3}}{(S')^{4/3}}. \quad (8.80)$$

Equation (8.79) describes a boundary layer problem having complicated behavior in the inner $|\xi| < 1$ region in the immediate vicinity of the hybrid resonance and, presumably, simple WKB-like behavior in the outer $|\xi| \gg 1$ region far from the hybrid resonance. To solve this problem the technique discussed for the Airy equation will now be applied and generalized:

1. Laplace transform: While Eq. (8.79) cannot be Fourier analyzed in the ξ direction, it can be Laplace transformed giving

$$p^4 \tilde{\phi}(p) + p \tilde{\phi}(p) - \frac{d}{dp} \left[p^2 \tilde{\phi}(p) \right] + \mu \tilde{\phi}(p) = 0 \quad (8.81)$$

or equivalently

$$\frac{1}{[p^2 \tilde{\phi}(p)]} \frac{d}{dp} [p^2 \tilde{\phi}(p)] = p^2 + \frac{1}{p} + \frac{\mu}{p^2}. \quad (8.82)$$

This has the solution

$$\tilde{\phi}(p) = A \exp(p^3/3 - \mu/p - \ln p), \quad (8.83)$$

where A is a constant. The inverse transform is

$$\phi(\xi) = A \int_C e^{f(p)} dp, \quad (8.84)$$

where

$$f(p) = \frac{p^3}{3} - \frac{\mu}{p} - \ln(p) + p\xi. \quad (8.85)$$

2. Boundary conditions: Since Eq. (8.79) is a fourth-order ordinary differential equation, *four* independent solutions must exist with four associated independent choices for the contour C in Eq. (8.84). The appropriate contour is determined by the imposed physical boundary conditions, which must be equal in number to the order of the equation. We consider a specific physical problem where an external antenna located at $\xi \gg 0$ generates a cold plasma wave propagating to the left. This physical prescription imposes the following four boundary conditions:

- (a) To the right of $\xi = 0$ there is a cold plasma wave with *energy* propagating to the left (i.e., towards the hybrid layer). It is important to recall that, as demonstrated in Eq. (6.105), the group velocity for these waves is orthogonal to the phase velocity so the wave is backwards in either the x or the z direction. The dispersion of these waves is given in Eq. (6.102) and the group velocity is given in Eq. (6.104). A plot of $S(\omega)$ and $P(\omega)$ shows that $\partial S/\partial \omega > 0$ and $\partial P/\partial \omega > 0$. Thus, for the situation where $S > 0$ and $P < 0$, the x components of the wave phase and group velocities have opposite signs. Hence, the cold wave phase velocity must propagate to the *right* to be consistent with the boundary condition that cold wave energy is propagating to the *left*.
- (b) To the right of $\xi = 0$ there is *no* warm plasma wave propagating to the left (for the hot wave, energy and phase propagate in the same direction).
- (c) To the left of $\xi = 0$ the cold plasma evanescent mode vanishes as $\xi \rightarrow -\infty$.
- (d) To the left of $\xi = 0$ the hot plasma evanescent mode vanishes as $\xi \rightarrow -\infty$.

The question is: what happens at the hybrid layer? Possibilities include absorption of the incoming cold plasma wave (unlikely since there is no dissipation in this problem), reflection of the incoming cold plasma wave at the hybrid layer, or mode conversion.

3. Calculation of saddle points: The saddle points are the roots of

$$f'(p) = p^2 + \xi + \frac{\mu}{p^2} - \frac{1}{p} = 0. \quad (8.86)$$

For large ξ , these roots separate into two large roots (hot mode), which are obtained when the first two terms in Eq. (8.86) balance each other and two small roots (cold

mode), which are obtained when the second and third terms balance each other. The large roots satisfy $p^2 = -\xi$ and the small roots satisfy $p^2 = -\mu/\xi$. For large ξ , the fourth term is small compared to the dominant terms for both large and small roots. The quantity f can be approximated for large p as

$$\begin{aligned} f(p) &\simeq \frac{p^3}{3} + p\xi - \ln p \\ &= p\left(\frac{p^2}{3} + \xi\right) - \ln p \\ &= \frac{2}{3}p\xi - \ln p \end{aligned} \quad (8.87)$$

while for small p it can be approximated as

$$\begin{aligned} f(p) &\simeq p\xi - \frac{\mu}{p} - \ln p \\ &= p\left(\xi - \frac{\mu}{p^2}\right) - \ln p \\ &= 2p\xi - \ln p. \end{aligned} \quad (8.88)$$

For $\xi > 0$ and large roots, the quantities p_l , $f(p_l)$, and $f''(p_l)$ are

$$p_l = \pm(-\xi)^{1/2}, \quad f(p_l) = \mp \frac{2}{3}(-\xi)^{3/2} - \ln p_l, \quad f''(p_l) = \pm 2(-\xi)^{1/2}. \quad (8.89)$$

For $\xi > 0$ the corresponding quantities for the small roots are

$$p_s = \pm(-\mu/\xi)^{1/2}, \quad f(p_s) = \pm 2(-\xi\mu)^{1/2} - \ln p_s, \quad f''(p_s) = \mp 2 \frac{(-\xi)^{3/2}}{\mu^{1/2}}. \quad (8.90)$$

The Gaussian integrals corresponding to steepest descent paths over these $\xi > 0$ saddle points are

$$\text{large roots: } \int_{\text{vicinity of saddle}} e^{f(p)} dp = \sqrt{\frac{\mp \pi}{(-\xi)^{3/2}}} e^{\mp \frac{2}{3}(-\xi)^{3/2}} \quad (8.91)$$

$$\text{small roots: } \int_{\text{vicinity of saddle}} e^{f(p)} dp = \sqrt{\frac{\pm \pi}{(-\mu\xi)^{1/2}}} e^{\pm 2(-\xi\mu)^{1/2}}, \quad (8.92)$$

where the logarithmic term in $f(p)$ has been taken into account. For $\xi < 0$, the large root quantities are

$$p_l = \pm|\xi|^{1/2}, \quad f(p_l) = \mp \frac{2}{3}|\xi|^{3/2} - \ln p_l, \quad f''(p_l) = \pm 2|\xi|^{1/2} \quad (8.93)$$

and the small root quantities are

$$p_s = \pm(\mu/|\xi|)^{1/2}, \quad f(p_s) = \mp 2(|\xi|\mu)^{1/2} - \ln p_s, \quad f''(p_s) = \mp 2 \frac{|\xi|^{3/2}}{\mu^{1/2}}. \quad (8.94)$$

4. Choice of contour path: For $\xi < 0$ and assuming $\mu > 0$, the saddle points giving solutions that vanish when $\xi \rightarrow -\infty$ are the saddle points with the upper (i.e., minus) sign chosen in the argument of the exponential. Hence, for $\xi < 0$ a contour must be chosen that passes through one or both of these saddle points. For $\xi > 0$ the upper sign corresponds to waves propagating to the left (i.e., towards the hybrid layer). Hence, we allow the upper sign (minus) for the small root (i.e., cold mode), but not for the large root since one of the boundary conditions was that there is no inward propagating hot plasma wave. To proceed further, it is necessary to look at the topography of the real part of $f(p)$ for both signs of ξ ; a method for plotting this topography is the subject of Assignment 2.

From these plots it is seen that the correct joining is given by

$$\left\{ \begin{array}{l} i \frac{\exp(-\frac{2}{3}|\xi|^{3/2})}{|\xi|^{3/4}} + \\ \frac{\exp[-2\mu|\xi|^{1/2}]}{(|\xi|\mu)^{1/4}} \end{array} \right\} \longleftrightarrow \left\{ \begin{array}{l} \frac{\exp[\frac{2}{3}(i\xi)^{3/2}]}{i^{1/2}\xi^{3/4}} + \\ i^{1/2} \frac{\exp[2i(\mu\xi)^{1/2}]}{(\xi\mu)^{1/4}} \end{array} \right\}, \quad (8.95)$$

evanescent side, $\xi < 0$ propagating side, $\xi > 0$

which shows that a cold wave with energy propagating into the $S = 0$ layer is converted into a hot wave that propagates back out.

5. WKB connection: The quantities in Eq. (8.95) can be expressed as integrals having the same form as WKB solutions. For example, the cold propagating term can be written as

$$\begin{aligned} \frac{\exp[2i(\mu\xi)^{1/2}]}{(\xi\mu)^{1/4}} &= \sqrt{2} \frac{\exp[i \int_0^\xi (\mu/\xi')^{1/2} d\xi']}{\sqrt{\int_0^\xi (\mu/\xi')^{1/2} d\xi'}} \\ &= \sqrt{2} \frac{\exp[i \int_0^x (-k_z^2 P/S)^{1/2} dx']}{\sqrt{\int_0^x (-k_z^2 P/S)^{1/2} dx'}}, \end{aligned} \quad (8.96)$$

where the last form is clearly the WKB solution. A similar identification exists for the hot plasma mode, so that Eq. (8.95) can also be written in terms of the WKB solutions.

This is just one example of mode conversion; other forms occur in different contexts but a similar analysis may be used and similar joining conditions are obtained. A curious feature of the mode conversion analysis is that the differential equation is never explicitly solved near $\xi = 0$; all that is done is match asymptotic solutions for the region where ξ is large and positive to the solutions for the region where ξ is large and negative.

8.6 Drift waves

Only textbook plasmas are uniform – real plasmas have both finite extent and gradients in various parameters such as pressure, magnetic field, etc. A finite-extent, magnetically confined warm plasma necessarily has a pressure gradient perpendicular to the magnetic field. For example, consider an azimuthally symmetric cylindrical plasma immersed in a strong axial magnetic field as sketched in Fig. 8.3; the pressure is assumed to be peaked on the z axis and to fall off radially. Particles stream freely in the axial direction but are constrained to make Larmor orbits in the perpendicular direction. Since the concept of pressure gradient has no meaning for an individual particle, consideration of the effect of pressure gradients requires a fluid or Vlasov point of view. This will be done first using a two-fluid analysis, then using a Vlasov analysis.

From the two-fluid point of view, the radial pressure gradient implies an equilibrium force balance

$$0 = q_\sigma \mathbf{u}_\sigma \times \mathbf{B} - n_\sigma^{-1} \nabla (n_\sigma \kappa T_\sigma). \quad (8.97)$$

Solving Eq. (8.97) for $\mathbf{u}_\sigma$ shows that each species has a steady-state perpendicular motion at the *diamagnetic drift velocity*

$$\mathbf{u}_{d\sigma} = - \frac{\nabla (n_\sigma \kappa T_\sigma) \times \mathbf{B}}{q_\sigma n_\sigma B^2}, \quad (8.98)$$

which is in the azimuthal direction. The corresponding diamagnetic drift current is

$$\begin{aligned} \mathbf{J}_d &= \sum_\sigma n_\sigma q_\sigma \mathbf{u}_{d\sigma} \\ &= - \frac{1}{B^2} \sum_\sigma \nabla (n_\sigma \kappa T_\sigma) \times \mathbf{B} \\ &= - \frac{1}{B^2} \nabla P \times \mathbf{B}, \end{aligned} \quad (8.99)$$

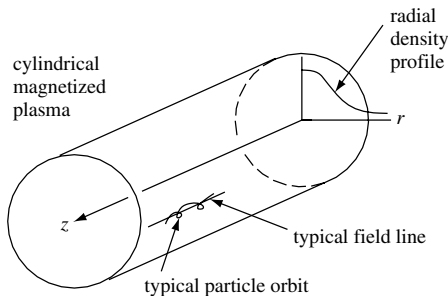


Fig. 8.3 Cylindrical magnetized plasma with radial density gradient.

which is just the azimuthal current associated with the MHD equilibrium equation $\mathbf{J} \times \mathbf{B} = \nabla P$, where $P = P_i + P_e$.

The electron and ion diamagnetic drift velocities thus provide the current necessary to establish the magnetic force that balances the MHD pressure gradient. It turns out that magnetized plasmas with pressure gradients are unstable to a class of electrostatic modes called “drift” waves. These modes exist in the same frequency regime as MHD but do not appear in standard MHD models, since standard MHD models do not provide a sufficiently detailed characterization of the differences between electron and ion dynamics.

From the particle point of view, the diamagnetic velocity is entirely fictitious because no particle actually moves at this velocity. The diamagnetic drift velocity is nevertheless quite genuine from the fluid point of view and so a fluid-based wave analysis must linearize about an equilibrium that includes this equilibrium drift. As will be seen later, the Vlasov point of view confirms and extends the conclusions of the fluid theory, providing care is taken to use an equilibrium velocity distribution function that is both valid and consistent with existence of a pressure gradient.

Drift waves will be examined using three progressively more realistic points of view. These are:

1. A collisionless two-fluid model where ions are assumed to be cold and electrons are assumed to be hot. This model establishes existence of the mode and provides a derivation for the intrinsic frequency, but provides no information regarding stability.
2. A collisional two-fluid model, which shows that *collisions* destabilize drift waves.
3. A Vlasov model including both finite ion temperature and net axial current. This model shows that both Landau damping and axial currents destabilize drift waves.

Drift waves involve physically distinct ion and electron motions, three-dimensional geometry, magnetized warm plasma effects, pressure gradients, collisionality, and Landau damping/instability. These waves are often of practical importance because they are so easily driven unstable. While real plasmas have geometry resembling Fig. 8.3, the cylindrical geometry will now be replaced by Cartesian geometry to simplify the analysis.

Using Cartesian geometry, the equilibrium magnetic field is assumed to be in the z direction and the pressure gradient is accounted for by assuming an exponential density gradient in the x direction. The x direction thus corresponds to the r direction of cylindrical geometry, and the y direction corresponds to the θ direction. The y and z directions, but not the x direction, are ignorable coordinates, and so the governing equations may be Fourier-transformed in the y and z directions, but not in the x direction.

8.6.1 Simple two-fluid model of drift waves

The plasma is assumed to have the exponential density gradient

$$n \sim \exp(-x/L), \quad (8.100)$$

where L is called the density-gradient scale length. We consider electrostatic, extremely low-frequency ($\omega \ll \omega_{ci}$) waves having the potential perturbation

$$\phi_1 = \tilde{\phi} \exp(ik_y y + ik_z z - i\omega t). \quad (8.101)$$

The parallel phase velocity is assumed to lie in the range $v_{Ti} \ll \omega/k_z \ll v_{Te}$ so that ions are adiabatic and electrons are isothermal (this is the same regime as ion acoustic waves). The parallel and perpendicular wavelengths are both assumed to be much longer than the electron Debye length, so the plasma may be considered quasi-neutral and have $n_i \simeq n_e$. Since $\omega/k_z \ll v_{Te}$, the parallel component of the electron equation of motion is simply

$$0 \simeq -q_e \frac{\partial \phi_1}{\partial z} - \frac{1}{n_e} \frac{\partial}{\partial z} (n_e \kappa T_e), \quad (8.102)$$

which leads to a Boltzmann electron density,

$$n_e = n_{e0} \exp(-q_e \phi_1 / \kappa T_e). \quad (8.103)$$

Assuming ϕ_1 is small, linearization of this Boltzmann electron density gives the first-order electron density

$$\frac{n_{e1}}{n_{e0}} = -\frac{q_e \phi_1}{\kappa T_e}. \quad (8.104)$$

Since $\omega/k_z \gg v_{Ti}$, ions may be considered cold to first approximation and the zero-pressure limit of the ion equation of motion characterizes ion dynamics. Furthermore, since $\omega \ll \omega_{ci}$ the lowest-order perpendicular ion motion is just the $\mathbf{E} \times \mathbf{B}$ drift so

$$\mathbf{u}_{i1} = \frac{-\nabla \phi_1 \times \mathbf{B}}{B^2} = -\frac{ik_y \phi_1}{B} \hat{x}. \quad (8.105)$$

Since the $\mathbf{E} \times \mathbf{B}$ drift is in the x direction, this drift is *in the direction of the density gradient* and leads to an ion density perturbation because of a convective interaction with the equilibrium density gradient. The ion density perturbation is found by linearizing the ion continuity equation

$$\frac{\partial n_{i1}}{\partial t} + \mathbf{u}_{i1} \cdot \nabla n_{i0} + n_{i1} \nabla \cdot \mathbf{u}_{i1} = 0. \quad (8.106)$$

After noting that $\nabla \cdot \mathbf{u}_{i1} = 0$, substitution for $\mathbf{u}_{i1}$ in the convective term gives

$$\frac{n_{i1}}{n_{i0}} = \frac{k_y \phi_1}{\omega L B}. \quad (8.107)$$

The density perturbation results from the ion $\mathbf{E} \times \mathbf{B}$ drift causing the ions with their density gradient to slosh back and forth in the x direction so that a stationary observer at a fixed point x sees oscillations of the ion density.

Quasi-neutrality means that the electron and ion density perturbations must be almost exactly equal although electrons and ions are governed by entirely different physics. Equating the electron and ion perturbation densities, respectively given by Eqs. (8.104) and (8.107), provides the most basic form of the drift wave dispersion relation,

$$\omega = -\frac{k_y \kappa T_e}{q_e L B} = k_y u_{de}, \quad (8.108)$$

where

$$u_{de} = -\kappa T_e / q_e L B \quad (8.109)$$

is the electron diamagnetic drift velocity given by Eq. (8.98). Equation (8.108) describes a normal mode where Boltzmann electron density perturbations caused by isothermal electron motion along field lines neutralize ion density perturbations caused by ion $\mathbf{E} \times \mathbf{B}$ motion sloshing the x -dependent equilibrium density profile in the x direction. This basic dispersion relation provides no information about the mode stability because collisions and Landau damping have not yet been considered. The wave phase velocity ω/k_y is equal to the electron diamagnetic drift velocity given by Eq. (8.98). The basic drift wave dispersion relation has the interesting feature that it does not depend on the mass of either species; this is because neither the electron Boltzmann dependence nor the ion $\mathbf{E} \times \mathbf{B}$ drift depend on mass. This basic dispersion relation provides a foundation for the more complicated models to be considered in the next two sections, and it is conventional to define the “drift frequency”

$$\omega^* = k_y u_{de}, \quad (8.110)$$

which will appear repeatedly in the more complicated models. The basic dispersion is therefore simply $\omega = \omega^*$ (the asterisk should not be confused with complex conjugate).

8.6.2 Two-fluid drift wave model with collisions

The next level of sophistication involves assuming the plasma is mildly collisional so that the electron equation of motion is now

$$m_e \frac{d\mathbf{u}_e}{dt} = q_e (-\nabla\phi + \mathbf{u}_e \times \mathbf{B}) - \frac{1}{n_e} \nabla (n_e \kappa T_e) - \nu_{ei} m_e (\mathbf{u}_e - \mathbf{u}_i). \quad (8.111)$$

The collision frequency ν_{ei} is assumed to be small compared to the electron cyclotron frequency $\omega_{ce} = q_e B / m_e$. Because $\omega/k_z \ll v_{Te}$, the electron inertia term

on the left-hand side is negligible compared to the electron pressure gradient term on the right-hand side and so the parallel component of the electron equation reduces to

$$0 = -q_e \frac{\partial \phi}{\partial z} - \frac{\kappa T_e}{n_e} \frac{\partial n_e}{\partial z} - \nu_{ei} m_e u_{ez}; \quad (8.112)$$

the ion parallel velocity has been dropped since it is negligible compared to the electron parallel velocity. After linearization and assuming perturbed quantities have a space-time dependence given by Eq. (8.101), Eq. (8.112) can be solved to give the parallel electron velocity

$$u_{ez} = -\frac{ik_z \kappa T_e}{\nu_{ei} m_e} \left(\frac{q_e \phi_1}{\kappa T_e} + \frac{n_{e1}}{n_{e0}} \right). \quad (8.113)$$

It is seen that in the limit of no collisions, this reverts to the Boltzmann result, Eq. (8.104).

As in the previous section, quasi-neutrality is invoked so the dispersion relation is obtained by equating the perturbed electron and ion densities. Equation (8.113) together with the continuity equation gives the perturbed electron density, provided the electron perpendicular velocity is also known. Since it has been assumed that $|\omega_{ce}| \gg \nu_{ei}$, the magnetic force term in Eq. (8.111) is much greater than the perpendicular component of the collision term. The perpendicular component of the electron equation of motion is therefore

$$0 = q_e (-\nabla_{\perp} \phi + \mathbf{u}_e \times \mathbf{B}) - \frac{1}{n_e} \nabla_{\perp} (n_e \kappa T_e). \quad (8.114)$$

In previous derivations the continuity equation was typically linearized right away, but here it turns out to be computationally advantageous to postpone linearization and instead solve Eq. (8.114) as it stands for the perpendicular electron flux to obtain

$$\Gamma_{e\perp} = n_e \mathbf{u}_{e\perp} = -\frac{n_e \nabla \phi \times \mathbf{B}}{B^2} - \frac{1}{q_e B^2} \nabla (n_e \kappa T_e) \times \mathbf{B}. \quad (8.115)$$

Using the vector identities $\nabla \cdot (\nabla \phi \times \mathbf{B}) = 0$ and $\nabla \cdot [\nabla (n_e \kappa T_e) \times \mathbf{B}] = 0$ it is seen that the divergence of the perpendicular electron flux is

$$\nabla \cdot \Gamma_{e\perp} = -\nabla n_e \cdot \frac{\nabla \phi \times \mathbf{B}}{B^2}. \quad (8.116)$$

The electron continuity equation can be expressed as

$$\frac{\partial n_e}{\partial t} + \frac{\partial}{\partial z} (n_e u_{ez}) + \nabla \cdot \Gamma_{e\perp} = 0, \quad (8.117)$$

and, after substitution for the perpendicular electron flux, this becomes

$$\frac{\partial n_e}{\partial t} + \frac{\partial}{\partial z}(n_e u_{ez}) - \nabla n_e \cdot \frac{\nabla \phi \times \mathbf{B}}{B^2} = 0. \quad (8.118)$$

At this stage of the derivation it is now appropriate to linearize the continuity equation, which becomes

$$\frac{\partial n_e}{\partial t} + n_{e0} \frac{\partial u_{ez1}}{\partial z} - \frac{dn_{e0}}{dx} \hat{x} \cdot \frac{\nabla \phi_1 \times \mathbf{B}}{B^2} = 0. \quad (8.119)$$

Substituting for ϕ_1 using Eq. (8.101), and using Eq. (8.113) for u_{z1} gives

$$\frac{n_{e1}}{n_{e0}} \left(-i\omega + \frac{k_z^2 \kappa T_e}{\nu_{ei} m_e} \right) + \frac{q_e \phi_1}{\kappa T_e} \left(\frac{k_z^2 \kappa T_e}{\nu_{ei} m_e} + \frac{ik_y \kappa T_e}{q_e B L} \right) = 0, \quad (8.120)$$

which may be solved to give the sought-after electron density perturbation

$$\frac{n_{e1}}{n_{e0}} = -\frac{q_e \phi_1}{\kappa T_e} \frac{\left(-i\omega^* + \frac{1}{\tau_{\parallel}} \right)}{\left(-i\omega + \frac{1}{\tau_{\parallel}} \right)}. \quad (8.121)$$

Here

$$\tau_{\parallel} = \nu_{ei} m_e / k_z^2 \kappa T_e \quad (8.122)$$

is the nominal time required for electrons to diffuse a distance of the order of a parallel wavelength. This is because the parallel collisional diffusion coefficient scales as $D_{\parallel} \sim (\text{random step})^2 / (\text{collision period}) \sim \kappa T_e / m_e \nu_{ei}$ and, according to the diffusion equation, the distance diffused in time t is given by $z^2 \sim 4D_{\parallel} t$. Equation (8.121) shows that in the limit $\tau_{\parallel} \rightarrow 0$, the electron density perturbation reverts to being Boltzmann but, for finite $\tau_{\parallel}$, a phase lag occurs between n_{e1} and ϕ_1 .

The next step is to calculate the ion density perturbation. The ion equation of motion is

$$m_i \frac{d\mathbf{u}_i}{dt} = q_i (-\nabla \phi + \mathbf{u}_i \times \mathbf{B}) - \nu_{ie} m_i (\mathbf{u}_i - \mathbf{u}_e); \quad (8.123)$$

because the ions are heavy, the left-hand side inertial term must now be retained. The ions are assumed to be cold and the ion inertial term is assumed to be much larger than the collisional term so the parallel component of the linearized Eq. (8.123) is

$$u_{iz1} = \frac{k_z q_i \phi_1}{\omega m_i} = \frac{k_z c_s^2}{\omega} \frac{q_i \phi_1}{\kappa T_e}, \quad (8.124)$$

where $c_s^2 = \kappa T_e / m_i$ is the ion acoustic velocity. The appearance of the ion acoustic velocity comes from normalizing the potential perturbation to κT_e ; this normalization is used in order to be consistent with the normalization used for the electron dynamics. Noting again that the inertial term is assumed to be much larger than the collisional term, the linearized perpendicular ion equation becomes

$$m_i \frac{d\mathbf{u}_{i\perp}}{dt} = q_i (-\nabla\phi_1 + \mathbf{u}_{i\perp} \times \mathbf{B}), \quad (8.125)$$

which can be solved to give the perpendicular velocity as a sum of an $\mathbf{E} \times \mathbf{B}$ drift and a polarization drift, i.e.,

$$\mathbf{u}_{i\perp} = \frac{-\nabla\phi \times \mathbf{B}}{B^2} + \frac{m_i}{q_i B^2} \frac{d}{dt} (-\nabla\phi). \quad (8.126)$$

The polarization drift is retained in the ion but not the electron equation, because polarization drift is proportional to mass. The perpendicular ion flux is

$$\Gamma_{i\perp} = \frac{-n_i \nabla\phi \times \mathbf{B}}{B^2} + \frac{n_i m_i}{q_i B^2} \frac{d}{dt} (-\nabla_\perp \phi), \quad (8.127)$$

which has a divergence

$$\nabla \cdot \Gamma_{i\perp} = -\nabla n_i \cdot \frac{\nabla\phi \times \mathbf{B}}{B^2} - \nabla \cdot \left(\frac{n_i m_i}{q_i B^2} \frac{d}{dt} (\nabla_\perp \phi) \right). \quad (8.128)$$

Substitution of Eqs. (8.128) and (8.124) into the ion continuity equation, linearizing, and solving for the ion density perturbation gives

$$\begin{aligned} \frac{n_{i1}}{n_{i0}} &= \left[\frac{k_z^2 c_s^2}{\omega^2} + \frac{q_e}{q_i} \frac{k_y \kappa T_e}{\omega q_e L B} - \frac{m_i \kappa T_e}{q_i^2 B^2} k_y^2 \right] \frac{q_i \phi_1}{\kappa T_e} \\ &= \left[\frac{k_z^2 c_s^2}{\omega^2} - \frac{q_e}{q_i} \frac{\omega^*}{\omega} - k_y^2 \rho_s^2 \right] \frac{q_i \phi_1}{\kappa T_e}, \end{aligned} \quad (8.129)$$

where $\rho_s^2 \equiv \kappa T_e / m_i \omega_{ci}^2$ is a fictitious ion Larmor orbit defined using the electron temperature instead of the ion temperature. The fictitious length ρ_s^2 is analogous to the fictitious velocity $c_s^2 = \kappa T_e / m_i$, which appeared in the analysis of ion acoustic waves.

Since the plasma is quasi-neutral, the normalized electron and ion density perturbations must be the same. Equating the normalized density perturbations obtained from Eq. (8.121) and (8.129) gives

$$\frac{\left(-i\omega^* + \frac{1}{\tau_{\parallel}}\right)}{\left(-i\omega + \frac{1}{\tau_{\parallel}}\right)} = \frac{\omega^*}{\omega} + \frac{k_z^2 c_s^2}{\omega^2} - k_y^2 \rho_s^2. \quad (8.130)$$

For simplicity it is assumed here that $q_i/q_e = -1$, which is the situation for low-temperature plasmas where the ions are too cold to be multiply ionized ($q_i/q_e = -1$ is, of course, always true for hydrogen plasmas). It is also assumed that the collision frequency is sufficiently small to have $\omega\tau_{\parallel} = \omega\nu_{ei}m_e/k_z^2 T_e \ll 1$ and that ω is of the order of ω^* ; the self-consistency of these assumptions will be checked later. The left-hand side of Eq. (8.130) can now be expanded using the binomial theorem to obtain the collisional drift wave dispersion relation

$$D(\omega, k_y, k_z) = 1 - \frac{\omega^*}{\omega} + k_y^2 \rho_s^2 - \frac{k_z^2 c_s^2}{\omega^2} + i(\omega - \omega^*)\tau_{\parallel} = 0. \quad (8.131)$$

This dispersion relation shows drift waves have an association with ion acoustic waves since in the limit $k_y^2 \rho_s^2 \rightarrow 0$, the real part of the dispersion becomes

$$1 - \frac{\omega^*}{\omega} - \frac{k_z^2 c_s^2}{\omega^2} = 0; \quad (8.132)$$

this dispersion relation reduces to an ion acoustic wave dispersion relation in the limit of there being no equilibrium pressure gradient (i.e., when ω^* vanishes). Equation (8.132) has two roots

$$\omega = \frac{\omega^* \pm \sqrt{(\omega^*)^2 + 4k_z^2 c_s^2}}{2}, \quad (8.133)$$

which for small $k_z c_s$ are

1. $\omega = \omega^*$ and $\omega = -|k_z c_s|$ if $\omega^* > 0$
2. $\omega = \omega^*$ and $\omega = |k_z c_s|$ if $\omega^* < 0$.

Thus, the drift mode is a distinct mode compared to the ion acoustic wave and its parallel phase velocity is much faster than the ion acoustic wave since the drift wave occurs in the limit $\omega/k_z \gg c_s$.

We now address the important question of the stability properties of Eq. (8.131). To do this, k_z is assumed to be sufficiently small to have $k_z c_s \ll \omega$ so the

dispersion describes the drift mode and not the ion acoustic mode. The real part of the dispersion is then set to zero to obtain the real part of the frequency

$$\omega_r = \frac{\omega^*}{1 + k_y^2 \rho_s^2} \quad (8.134)$$

showing that the actual frequency of the collisional drift wave is smaller than ω^* , but is of the same order, provided $k_y^2 \rho_s^2$ is not larger than order unity. The $k_y^2 \rho_s^2$ dependence results from ion polarization drift, an effect that was neglected in the initial simple model. The assumption $\omega \tau_{\parallel} = \omega \nu_{ei} m_e / k_z^2 T_e \ll 1$ implies $k_y \nu_{ei} / k_z^2 L \omega_{ce} \ll 1$, which is true provided the ratio ν_{ei} / ω_{ce} is sufficiently small for a given geometric factor $k_y / k_z^2 L$. Using the Taylor expansion technique discussed in the treatment of Eqs. (5.85)–(5.87), the imaginary part of the frequency is found to be

$$\omega_i = -\frac{D_i}{(\partial D_r / \partial \omega)_{\omega=\omega_r}} = -\frac{(\omega - \omega^*) \tau_{\parallel}}{\omega^* / \omega_r^2} = \omega^* \frac{k_y^2 \rho_s^2 \omega^* \tau_{\parallel}}{(1 + k_y^2 \rho_s^2)^3}. \quad (8.135)$$

Equation (8.135) shows ω_i has several important features:

1. ω_i is positive so collisional drift waves are *always unstable*.
2. ω_i is proportional to $\tau_{\parallel}$ so modes with the smallest k_z and hence the fastest parallel phase velocities are the most unstable, subject to the proviso that $\omega \tau_{\parallel} \ll 1$ is maintained. This increase of growth rate with k_z^{-1} (i.e., increase with parallel wavelength) means drift waves typically have the longest possible parallel wavelength allowed by the boundary conditions. For example, a linear plasma of finite axial extent with grounded conducting end walls has the boundary condition $\phi_1 = 0$ at both end walls; the longest allowed parallel wavenumber is π/h , where h is the axial length of the plasma (i.e., half a wavelength is the minimum number of waves that can be fitted subject to the boundary condition).
3. Collisions make the wave unstable since $\tau_{\parallel} \sim \nu_{ei}$.
4. ω_i is proportional to the factor $k_y^2 \rho_s^2 / (1 + k_y^2 \rho_s^2)^3$, which has a maximum when $k_y \rho_s$ is of order unity.
5. ω_i is proportional to L^{-2} ; for a realistic cylindrical plasma (cf. Fig. 8.3) the density is uniform near the axis and has a gradient near the edge (e.g., a Gaussian density profile where $n(r) \sim \exp(-r^2/L^2)$). The density gradient is localized near the edge of the plasma and so drift waves will have the largest growth rate near the edge. Drift waves in real plasmas are typically observed to have maximum amplitude in the region of maximum density gradient.

The free energy driving drift waves is the pressure gradient and so the drift waves might be expected to deplete their energy source eventually by flattening out the pressure gradient. This indeed happens and, in particular, it is the nonlinear behavior of drift waves that reduces the pressure gradient. This flattening is

accomplished by the drift waves pumping plasma from regions of high pressure to regions of low pressure. The pumping can be calculated by considering the time-averaged, nonlinear, x -directed particle flux associated with drift waves, namely

$$\Gamma_x = \langle \text{Re } n_1 \text{Re } u_{1x} \rangle = \frac{1}{2} \text{Re } (n_1 u_{1x}^*). \quad (8.136)$$

The species subscript σ has been omitted here, because to lowest order both species have the same u_{1x} given by the $\mathbf{E} \times \mathbf{B}$ drift, i.e.,

$$u_{1x} = \hat{x} \cdot \frac{-\nabla \phi_1 \times \mathbf{B}}{B^2} = -ik_y \phi_1 / B. \quad (8.137)$$

Being careful to remember that ω^* is a real quantity (this is a conventional, but confusing, notation), Eqs. (8.121) and (8.137) are substituted into Eq. (8.136) to obtain the wave-induced particle flux

$$\begin{aligned} \Gamma_x &= \frac{1}{2} \text{Re} \left[-\frac{n_0 q_e}{\kappa T_e} (1 - i(\omega^* - \omega) \tau_{\parallel}) \frac{ik_y |\phi_1|^2}{B} \right] \\ &= -\frac{1}{2} \text{Re} \left[\frac{n_0 q_e}{\kappa T_e} (\omega^* - \omega) \tau_{\parallel} \frac{k_y |\phi_1|^2}{B} \right] \\ &= \left(\frac{k_y^2 \rho_s^2}{1 + k_y^2 \rho_s^2} \right) \frac{k_y^2 |\phi_1|^2}{2LB^2} \frac{\nu_{ei} n_0 m_e}{k_z^2 \kappa T_e} \\ &= -D_{nl} \frac{dn_0}{dx}. \end{aligned} \quad (8.138)$$

Thus, it is seen that collisional drift waves cause an outward diffusion of plasma characterized by the nonlinear wave-induced diffusion coefficient

$$D_{nl} = \left(\frac{k_y^2 \rho_s^2}{1 + k_y^2 \rho_s^2} \right) \frac{k_y^2 |\phi_1|^2}{2B^2} \frac{\nu_{ei} m_e}{k_z^2 \kappa T_e}. \quad (8.139)$$

This diffusion coefficient has a proportionality similar to ω_i . The drift wave thus has the following properties: (i) the pressure gradient provides free energy and also an equilibrium where drift waves are a normal mode, (ii) collisions allow the drift wave to feed on the available free energy and grow, (iii) nonlinear diffusion flattens the pressure gradient thereby depleting the free energy and also undermining the confining effect of the magnetic field. A nonlinear saturated amplitude can result if some external source continuously replenishes the pressure gradient.

It was shown earlier that the most unstable drift waves are those having $k_y \rho_s$ of order unity. This property can be adapted to the more physically realistic situation of a cylindrical plasma by realizing that in a cylindrical plasma k_y is replaced by m/r , where m is the azimuthal mode number. Periodicity in θ forces m to be

an integer. If the most unstable m is a small integer, then the drift waves tend to be coherent, but if m is a large integer, many azimuthal wavelengths fit into the circumference of the cylinder. In this case, the periodicity condition is only a weak constraint and the waves typically become turbulent. Large m corresponds to small ρ_s , which occurs when the magnetic field is strong. Thus, plasmas with strong magnetic fields tend to have turbulent, short perpendicular wavelength drift waves, whereas plasmas with weak magnetic fields have coherent, long perpendicular wavelength drift waves.

Drift wave turbulence has been found to be the main reason why the radial diffusion in tokamak magnetic plasma confinement devices is much worse than can be explained by simple random walks due to particle collisions. The importance of drift wave turbulence can be estimated by assuming $k_y^2 \rho_s^2 \sim 1$ and then taking the ratio of D_{nl} to the classical diffusion given by Eq. (2.114) to obtain

$$\frac{D_{nl}}{D_{classical}} \sim \frac{1}{k_z^2 \rho_s^2} \left(\frac{e|\phi_1|}{\kappa T_e} \right)^2. \quad (8.140)$$

In the situation where the magnetic field is strong so ρ_s is a microscopic length (this length is $\sim r_{Li}$ if $T_e \sim T_i$) then $k_z \rho_s$ will be very small since k_z^{-1} is of the order of the axial extent of the configuration. Thus, if the turbulence level is sufficiently strong to satisfy the very modest requirement $e|\phi_1|/\kappa T_e > k_z \rho_s$, plasma transport across the magnetic field will be mainly from diffusion caused by drift wave turbulence, not classical diffusion.

8.6.3 Vlasov theory of drift waves: collisionless drift waves

Drift waves also exist when the plasma is so hot that the collision frequency becomes insignificant and the plasma can be considered collisionless. The previous section showed that a collision-induced phase lag between the density and potential fluctuations produces a destabilizing imaginary term in the dispersion relation. The analysis of collisionless plasma waves showed that Landau damping also causes a phase shift between the density and potential fluctuations resulting in an imaginary term in the dispersion relation. The Vlasov analysis in Assignment 8 of Chapter 5 showed that a relative motion between electrons and ions (i.e., a net current) destabilizes ion acoustic waves, and so it is reasonable to expect that a similar current-driven destabilization might also occur for drift waves.

The Vlasov analysis of drift waves is less intuitive than the fluid analysis, but the reward for abstraction is a more profound model. As in the fluid analysis, the plasma is assumed to have a uniform equilibrium magnetic field $\mathbf{B} = B\hat{z}$, an x -directed density gradient given by Eq. (8.100), and a perturbed potential given by Eq. (8.101). The Vlasov analysis is in a sense simpler than the fluid

analysis, because the Vlasov analysis involves a modest extension of the warm plasma electrostatic Vlasov model discussed in Section 8.1. As was discussed in Section 8.1 the perturbed distribution function is evaluated by integrating along the unperturbed orbits, i.e.,

$$f_{\sigma 1}(\mathbf{x}, \mathbf{v}, t) = \int_{-\infty}^t dt' \left[\frac{q_{\sigma}}{m_{\sigma}} \nabla \phi_1 \cdot \frac{\partial f_{\sigma 0}}{\partial \mathbf{v}} \right]_{\mathbf{x}=\mathbf{x}(t'), \mathbf{v}=\mathbf{v}(t')}. \quad (8.141)$$

The new feature here is that the equilibrium distribution function must incorporate the assumed density gradient.

A first instinct would be to accomplish this by simply multiplying the Maxwellian distribution function of Section 8.1 by the factor $\exp(-x/L)$ so that the assumed equilibrium distribution function would be $f_{\sigma 0}(x, \mathbf{v}) = (\pi v_{T\sigma})^{-3/2} \exp(-v^2/v_{T\sigma}^2 - x/L)$. This approach turns out to be wrong because $f_{\sigma 0}(x, \mathbf{v}) = (\pi v_{T\sigma})^{-3/2} \exp(-v^2/v_{T\sigma}^2 - x/L)$ is *not* a function of the constants of the motion and so is not a solution of the equilibrium Vlasov equation.

What is needed is some constant of the motion that includes the parameter x . The equilibrium distribution function could then be constructed from this constant of the motion and arranged to have the desired x -dependence. The appropriate constant of the motion is the canonical momentum in the y direction, namely

$$P_y = m_{\sigma} v_y + q_{\sigma} A_y = m_{\sigma} v_y + q_{\sigma} B_z x = \left(\frac{v_y}{\omega_{c\sigma}} + x \right) q_{\sigma} B_z \quad (8.142)$$

since $B_z = \partial A_y / \partial x$. Multiplying the original Maxwellian by the factor $\exp[-(v_y/\omega_{c\sigma} + x)/L]$ produces the desired spatial dependence while simultaneously satisfying the requirement that the distribution function depends only on constants of the motion. Note that P_y is a constant of the motion because y is an ignorable coordinate.

As suggested above, it turns out that z -directed currents can also destabilize collisionless drift waves, much like the situation where currents provide free energy that destabilizes ion acoustic waves. Since it takes little additional effort to include this possibility, a z -directed current will also be assumed so that electrons and ions are assumed to have unequal mean velocities $u_{\sigma z}$. In a frame moving with velocity $u_{\sigma z}$ the species σ is thus assumed to have a distribution function $\sim \exp[-mv^2/2 - (v'_y/\omega_{c\sigma} + x)/L]$, where $\mathbf{v}' = \mathbf{v} - u_{\sigma z} \hat{z}$ is the velocity in the moving frame. This is a valid solution to the equilibrium Vlasov equation, since both the energy $mv^2/2$ and the y -direction canonical momentum $(v'_y/\omega_{c\sigma} + x)/L$ are constants of the motion. The appropriately normalized lab-frame distribution function is

$$f_{\sigma 0}(x, \mathbf{v}) = \frac{n_{0\sigma}}{\pi^{3/2} v_{T\sigma}^3} \exp \left[-(\mathbf{v} - u_{\sigma z} \hat{z})^2 / v_{T\sigma}^2 - (v_y/\omega_{c\sigma} + x)/L - \left(\frac{v_T}{2\omega_{c\sigma} L} \right)^2 \right], \quad (8.143)$$

where the normalization factor $\exp\left(-(v_T/2\omega_c L)^2\right)$ has been inserted so that the zeroth moment of $f_{\sigma 0}$ gives the density. The necessity of this factor is made evident by completing the squares for the velocities and writing Eq. (8.143) in the equivalent form

$$f(x, \mathbf{v}) = \frac{n_0}{\pi^{3/2} v_T^3} \exp \left(- \left[\frac{v_x^2 + \left(v_y + \frac{v_T^2}{2\omega_c L} \right)^2 + (v_z - u_{\sigma z})^2}{v_T^2} \right] - \frac{x}{L} \right). \quad (8.144)$$

The term $v_y/\omega_{c\sigma}$, which crept into Eq. (8.143) because of the seemingly abstract requirement of having the distribution function depend on the constant of the motion P_y , corresponds to the fluid theory equilibrium diamagnetic drift. This correspondence is easily seen by calculating the mean y-direction velocity (i.e., first moment of Eq. (8.143)) and finding that the existence of the $v_y/\omega_{c\sigma}$ term results in a fluid velocity u_y , which is precisely the diamagnetic velocity given by Eq. (8.98).

We now insert Eq. (8.143) in Eq. (8.141) to calculate the perturbed distribution function. The unperturbed particle orbits are the same as in Section 8.1 because the concepts of both density gradient and axial current result from averaging over a distribution of many particles and so have no meaning for an individual particle. The integration over unperturbed orbits is thus identical to that of Section 8.1, except that a different equilibrium distribution function is used.

The essence of the calculation is in the term

$$\begin{aligned} \nabla \phi_1 \cdot \frac{\partial f_{\sigma 0}}{\partial \mathbf{v}} &= i \phi_1 \mathbf{k} \cdot \frac{\partial f_{\sigma 0}}{\partial \mathbf{v}} \\ &= -i \tilde{\phi} e^{i \mathbf{k} \cdot \mathbf{x}(t) - i \omega t} \left[\frac{2 \mathbf{k} \cdot \mathbf{v}}{v_{T\sigma}^2} - \frac{2 k_z u_{\sigma z}}{v_{T\sigma}^2} + \frac{k_y}{\omega_{c\sigma} L} \right] f_{\sigma 0}, \end{aligned} \quad (8.145)$$

which causes Eq. (8.141) to become

$$\begin{aligned} f_{\sigma 1}(\mathbf{x}, \mathbf{v}, t) &= -\frac{q_\sigma \tilde{\phi} f_{\sigma 0}}{m_\sigma} \int_{-\infty}^t dt' e^{i \mathbf{k} \cdot \mathbf{x}(t') - i \omega t'} \left[\frac{2 i \mathbf{k} \cdot \mathbf{v}}{v_{T\sigma}^2} - \frac{2 i k_z u_{\sigma z}}{v_{T\sigma}^2} + \frac{i k_y}{\omega_{c\sigma} L} \right] \\ &= -\frac{q_\sigma \tilde{\phi} f_{\sigma 0}}{m_\sigma} \int_{-\infty}^t dt' \left\{ \frac{2 e^{-i \omega t'}}{v_{T\sigma}^2} \left[\frac{d}{dt'} e^{i \mathbf{k} \cdot \mathbf{x}(t')} \right] - \left(\frac{2 i k_z u_{\sigma z}}{v_{T\sigma}^2} - \frac{i k_y}{\omega_{c\sigma} L} \right) e^{i \mathbf{k} \cdot \mathbf{x}(t') - i \omega t'} \right\} \\ &= -\frac{2 q_\sigma \tilde{\phi} f_{\sigma 0}}{m_\sigma v_{T\sigma}^2} \left[e^{i \mathbf{k} \cdot \mathbf{x}(t) - i \omega t} + i \left(\omega - k_z u_{\sigma z} - \omega_\sigma^* \right) \int_{-\infty}^t dt' e^{i \mathbf{k} \cdot \mathbf{x}(t') - i \omega t'} \right]; \end{aligned} \quad (8.146)$$

here

$$\frac{k_y v_{T\sigma}^2}{2\omega_{c\sigma} L} = \frac{k_y \kappa T}{qBL} = -\omega_{\sigma}^*, \quad (8.147)$$

has been used as a generalization of Eq. (8.110). The rest of the analysis is as before and gives the dispersion relation

$$1 + \sum_{\sigma} \frac{1}{k^2 \lambda_{D\sigma}^2} \left[1 + \frac{(\omega - k_z u_{\sigma z} - \omega_{\sigma}^*)}{k_z v_{T\sigma}} \sum_{n=-\infty}^{\infty} I_n(\Lambda_{\sigma}) e^{-\Lambda_{\sigma}} Z(\alpha_{n\sigma}) \right] = 0, \quad (8.148)$$

where, as before, $\Lambda_{\sigma} = k_{\perp}^2 r_{L\sigma}^2$ and $\alpha_{n\sigma} = (\omega - n\omega_{c\sigma})/k_z v_{T\sigma}$. It is seen that the density gradient and the axial current both provide a Doppler shift in the $\alpha_{0\sigma}$ term. In both cases the Doppler shift is given by the wavenumber in the appropriate direction times the mean fluid velocity in that direction.

Because Eq. (8.148) contains so much detailed physics in addition to the sought-after collisionless drift wave,¹ some effort and guidance is required to flush out the drift wave information from all the other information. This is done by taking appropriate asymptotic limits of Eq. (8.148) and guidance is obtained using the results from the two-fluid analysis. These results showed that drift waves exist in the regime where:

1. $v_{Ti} \ll \omega/k_z \ll v_{Te}$ so that the ions are adiabatic and the electrons are isothermal,
2. $\omega \sim \omega_e^* \ll \omega_{ci}, \omega_{ce}$,
3. $k_{\perp}^2 r_{Le}^2 \sim 0$ because the extremely small electron mass means that the electron Larmor orbit radius is negligible compared to the perpendicular wavelength even though the electrons are warm,
4. $k_{\perp}^2 r_{Li}^2 < 1$, since the ions were assumed cold,
5. $k_{\perp}^2 \rho_s^2 \sim 1$,
6. $k^2 \lambda_{De}^2 \ll 1$ so that the waves are quasi-neutral,
7. $|u_{zi}| \ll |u_{ze}|$, since the parallel equilibrium velocities satisfy $m_e u_{ez} + m_i u_{iz} = 0$; there is no parallel flow, just a parallel current.

Because $\Lambda_e \sim 0$, all electron terms in the summation over n vanish except for the $n = 0$ term. Since $\omega/k_z \ll v_{Te}$, the small-argument limit of the plasma dispersion function is used for electrons and it is seen that the imaginary term is the dominant term for the electrons. In contrast, since $v_{Ti} \ll \omega/k_z$, the large-argument limit of the plasma dispersion function is used for the ions and also equilibrium parallel motion is neglected for ions because of the large ion mass. For the $n \neq 0$ harmonics the terms $Z(\alpha_n) + Z(\alpha_{-n}) \rightarrow -k_z v_{Ti} [1/(\omega - n\omega_{ci}) + 1/(\omega + n\omega_{ci})]$

¹ Eq. (8.148) also describes Bernstein waves, the electrostatic limit of magnetized cold plasma waves, and mode conversion.

and hence cancel each other since $\omega \ll \omega_{ci}$. On making these approximations, Eq. (8.148) reduces to

$$1 + \frac{T_e}{T_i} \left[1 - e^{-\Lambda_i} I_0(\Lambda_i) \right] - \frac{\omega_e^*}{\omega} e^{-\Lambda_i} I_0(\Lambda_i) + i\pi^{1/2} \frac{(\omega - k_z u_{ze} - \omega_e^*)}{k_z v_{Te}} = 0. \quad (8.149)$$

The real part of this dispersion gives

$$\omega_r = \omega_e^* \left\{ \frac{e^{-\Lambda_i} I_0(\Lambda_i)}{1 + \frac{T_e}{T_i} [1 - e^{-\Lambda_i} I_0(\Lambda_i)]} \right\} \quad (8.150)$$

or in the limit of small Λ_i

$$\omega_r = \omega_e^* \left\{ \frac{1 - \Lambda_i}{1 + k_{\perp}^2 \rho_s^2} \right\}; \quad (8.151)$$

this corresponds to the fluid dispersion and moreover shows how finite ion temperature affects the dispersion. Using the method of Eq. (8.135), the imaginary part of the frequency is obtained from Eq. (8.149) as

$$\begin{aligned} \omega_i &= - \frac{\omega^2 \pi^{1/2}}{\omega_e^* e^{-\Lambda_i} I_0(\Lambda_i)} \frac{(\omega_r - k_z u_{ze0} - \omega_e^*)}{k_z v_{Te}} \\ &= \frac{\pi^{1/2} (\omega_e^*)^2}{k_z v_{Te}} \left\{ \frac{(1 - \Upsilon)(1 + T_e/T_i) + k_z u_{ze0} \Upsilon / \omega_r}{[1 + (1 - \Upsilon) T_e/T_i]^3} \right\}, \end{aligned} \quad (8.152)$$

where $\Upsilon = e^{-\Lambda_i} I_0(\Lambda_i)$. In the limit $\Lambda_i \rightarrow 0$, it is seen that $\Upsilon \rightarrow 1 - \Lambda_i$ so the imaginary part of the frequency becomes

$$\omega_i = \frac{\pi^{1/2} (\omega_e^*)^2}{k_z v_{Te} [1 + k_{\perp}^2 \rho_s^2]^3} \{ k_{\perp}^2 (\rho_s^2 + r_{Li}^2) + k_z u_{ze0} / \omega_r \}. \quad (8.153)$$

The imaginary part of the frequency is therefore always positive so there is always instability. Two collisionless destabilization mechanisms are seen to exist, *normal Landau damping*, represented by the term involving $k_{\perp}^2 (\rho_s^2 + r_{Li}^2)$ in the curly brackets, and *current*, represented by $k_z u_{ze0} / \omega_r$. Since collisions can also destabilize drift waves, there are at least three mechanisms by which drift waves can be destabilized. Comparison of Eq. (8.153) with Eq. (8.135) shows which of these mechanisms will be dominant in a given plasma. Modes with long parallel wavelengths are the most unstable for drift waves driven unstable by Landau damping, just as for collisional drift waves.

There are many more varieties of drift waves besides the basic versions discussed here. The common feature is that the gradient of pressure perpendicular to the magnetic field provides a new mode and free energy from the pressure

gradient or from parallel current can be tapped by this mode so the mode is spontaneously unstable.

8.7 Assignments

1. Electrostatic ion cyclotron waves. Using the electrostatic hot plasma dispersion relation show that there exists a mode in the vicinity of the ion cyclotron frequency having the dispersion

$$\omega^2 = \omega_{ci}^2 + k_{\perp}^2 c_s^2. \quad (8.154)$$

Show this mode can be driven unstable by an axial current.

2. Mode conversion. Work through the algebra of the linear mode conversion problem and plot the contours of $\text{Re } f(p)$ for both positive and negative ξ . Hint: it is easier to plot the contours of $f(p)$ if one uses polar coordinates in the complex p -plane so $p = re^{i\theta}$. Then the real part of the term p^3 is just $\text{Re } (r^3 e^{3i\theta}) = r^3 \cos(3\theta)$. Choose a set of contour paths for $\xi < 0$ that satisfy the boundary condition that there are no modes growing exponentially with increasing distance from $\xi = 0$. Then, determine which combination of these contours does not give an inward propagating hot plasma wave when $\xi > 0$ (this was another boundary condition).
3. Prove that the distribution function

$$f_{\sigma 0}(x, \mathbf{v}) = \frac{n_{0\sigma}}{\pi^{3/2} v_{T\sigma}^3} \exp \left[-(\mathbf{v} - u_{\sigma z} \hat{z})^2 / v_{T\sigma}^2 - (v_y / \omega_{c\sigma} + x) / L - \left(\frac{v_T}{2\omega_c L} \right)^2 \right]$$

provides a mean y -direction drift, which corresponds to the diamagnetic drift of fluid theory.